Program Information Form

Program Name	Bachelor Programme in Computer Engineering (2023 version) (%30 English)
Academic Unit	Department of Computer Engineering
Type	Undergraduate Major Program
Level Of Qualification	This is a First Cycle (Bachelor's Degree) Program
Qualification Awarded	The students who successfully complete the program are awarded the degree of Bachelor of Science (B.S.) in Bachelor Programme in Computer Engineering (2023 version) (%30 English)
Mode Of Study	Full-Time
Programme Director	M. Elif Karslıgil
Specific Admission Requirements	Those who want to enroll in YTU undergraduate degree programs must get the sufficient score required by YTU from the exam administered by the Student Selection and Placement Center (OSYM) and should not have an existing enrollment in another higher education program. The rules and regulations in "Directive On Application and Registration of Foreign Students" are applied to the students from abroad who want to enroll in this program. The students who qualify to enroll in undergraduate degree programs whose medium of instruction is 30% English have to take English Proficiency Exam. "Directive on Instruction and Examination, School of Foreign Languages (YDYO)-YTU" and other regulations apply to English Proficiency Exam and Preparatory English Courses, except for the Foreign Languages Department English Language Teaching Program.
Specific Arrangements For Recognition Of Prior Learning	Admissions to YTU undergraduate programs via transfers from outside YTU are conducted as per the principles determined by the Senate within the framework of the provisions of the Rules and Regulations on the Principles of Transfers between Associate and Bachelor's Degree Programs at Higher Education Institutions, Double Major, Double Minor and Credit Transfers Between Higher Education Institutions published in the Official Gazette No.27561 of 24/4/2010. Procedures for the students placed in this programs through the Vertical Transfer Exam held by the Student Selection and Placement Center (OSYM) are to be carried out pursuant to the rules and principles stipulated in the Regulation on Transfer of the Students graduated from Vocational Schools and Open University Associate Degrees to the Undergraduate Education published in the Official Gazette No.24676 of 19/2/2002.
Qualification Requirements And Regulations	The undergraduate students in this program must be successful in all of the courses with a minimum achievement grade of DC, must have completed at least 240 ECTS credits and have scored a minimum CGPA of 2.00/4.00. At the same time, the students must complete their compulsory internship within the designated period of time and within the scope of necessary qualifications.
Profile Of The Programme	The purpose of the Department of Computer Engineering is essentially to provide professional training in the world-class level and at the same time to participate actively in applied and theoretical research. So the graduated engineers have universal acceptance, solid foundation, ability for learning and leadership during their lifetimes and of course strong ethical values.
Occupational Profiles Of Graduates With Examples	Graduates of this program are expected to build their career on designing or research&development of computing and information systems in various sectors. They can work either in private or public companies in the field.
Access To Further Studies	The graduates of this program can apply to master programs to enhance their academic skills and career.

Examination Regulations Assessment And Grading	**Assessment of Success** a) In assessing a student’s performance in a course, the grade the student has scored during the semester work over a hundred and the grade the student has scored at the end of the semester over a hundred are taken into consideration. b) In measuring success, the weight of the grade during the semester is 60% and the weight of the final exam is 40%. **Achievement Grade** (1) In determining a grade, relative evaluation system is used. Achievement Grade is designated as follows: a) The meanings of the achievement grades are defined as follows: **Achievement Grade** / **Coefficient** / **Achievement Degree** AA / 4.00 / Excellent BA / 3.50 / Very good BB / 3.00 / Good CB / 2.50 / Average CC / 2.00 / Satisfactory DC / 1.50 / Provisionally Successful DD / 1.00 / Fail FD / 0.50 / Fail FF / 0.00 / Fail F0 / 0.00 / NA G: Pass K: Fail İ: Leave of Absence M: Exemption E: Incomplete 2) (DC) indicates that the student has been provisionally successful in a course. For a student to be considered successful in a course, he must have a minimum GPA of 2.0. If a student has courses in which he has been provisionally successful in his instructional plan, he must have a minimum GPA of 2.0 to qualify for graduation. And, this course is included in his GPA. 3) G (Pass) indicates that the student has been successful/satisfactory in a course and not included in his GPA. 4) K (Fail) indicates that the student has been unsuccessful/unsatisfactory in a course and not included in his GPA. 5) İ (Leave of Absence) indicates that the student has been unable to complete the requirements of a course because of sickness or some other valid reason

Achievement Grade	Coefficient	Achievement Degree
AA	4.00	Excellent

	pursuant to the relevant provision of this Regulation and is not included in GPA until it is transformed into an achievement grade. If this course is not completed the following semester in which the course is available, İ automatically turns into an FF. 6) M (Exemption) indicates that the student have exemption for the previous program courses which are deemed equivalent to the courses offered in their new undergraduate program. Decision for the course exemption is made by the relevant faculty committee. The courses that student is exempt from are processed as a non-credit exemption and they are not included in the student's GPA. **Make-up, Resit and Graduation Exams** (1) A make-up exam is administered in place of a mid-term exam. In case of multiple make-up exams, the student can only sit in one of these exams. The provisions stipulated by the Senate apply to whether a student can sit in a make-up exam or how to administer a make-up exam. A make-up exam for the exams at the end of the semester won't be allowed. (2) The provisions regarding resit exams are as follows: a) For a student to be able to sit in a resit exam, he must have added the course at the beginning of the semester and must have fulfilled the requirements to be able to take this exam at the end of the semester. Students who have missed a resit exam cannot have a make-up exam for it. b) Students who have been unsuccessful or provisionally successful (not F0) can sit in resit exams. The score in a resit exam is considered a final at the end of the semester. An achievement grade is assigned at the end of a resit exam by taking the percentages of visas, assignments and the resit exam into consideration. c) A student who have missed a resit exam gets E (Incomplete) and remains as the achievement grade of the course. The resit achievement grades are included in semester grade average points. (3) The provisions regarding graduation exams are as follows: a) To be able to sit in a graduation exam, a student must have fulfilled the requirements to take the final exam at the end of the semester. The students who haven't qualified for a graduation exam can't sit in a make-up exam for this exam. b) The students who have to pass a maximum of two courses before their graduation are granted a graduation exam for the classes they have failed after the resit exam and within the period stated in the academic calendar. The students who are unable to graduate due to their GPA below 2.00 can take a graduation exam in two courses in which they have been provisionally successful. c) To be considered successful in a graduation exam, a student must get at least a CC. The grade taken in the exam takes the place of the achievement grade of the course. Visas and assignments aren't included in the assessment.
Graduation Requirements	To be able to qualify for graduation, students must complete all the courses in the instructional plan, assignments, field work, applied projects, assignments, workshops, seminars, attendance, laboratory work and other related activities with a minimum CGPA of 2.00.

Program Outcomes	
1	PO-1.1) Getting sufficient knowledge in the field of Mathematics and Science
2	PO-1.2) Getting sufficient knowledge in the discipline-specific field
3	PO-1.3) An ability to apply knowledge gained in the discipline-specific field to solve complex engineering problems
4	PO-2.1) An ability to identify, define, formulate, and solve complex engineering problems
5	PO-2.2) An ability to select and apply appropriate analysis and modeling methods for this purpose.
6	PO-3.1) Ability to design a complex system, process, device or product to meet specific requirements under realistic constraints and conditions.
7	PO-3.2) An ability to apply modern design methods for this purpose.
8	PO-4.1) Ability to develop, select and use modern techniques and tools necessary for the analysis and solution of complex problems encountered in engineering applications.
9	PO-4.2) An ability to use information technologies effectively.
10	PO-5.1) An ability to design experiments to study complex engineering problems or discipline-specific research topics.
11	PO-5.2) Ability to conduct experiments to investigate complex engineering problems or discipline-specific research topics.
12	PO-5.3) Ability to collect data to examine complex engineering problems or discipline-specific research topics.
13	PO-5.4) An ability to analyze and interpet results for the study of complex engineering problems or discipline-specific research topics.
14	PO-6.1) Ability to work individually.
15	PO-6.2) Ability to work effectively in disciplinary teams.
16	PO-6.3) Ability to work effectively in multidisciplinary teams.
17	PO-7.1) An ability to communicate effectively, both orally and in writing.
18	PO-7.2) At least one foreign language knowledge.
19	PO-7.3) Ability to write effective reports and understand written reports.
20	PO-7.4) Ability to prepare design and production reports.
21	PO-7.5) Ability to give clear and understandable instructions.
22	PO-8.1) Being aware of the need for lifelong learning.
23	PO-8.2) Ability to constantly renew itself, access information and follow the developments in science and technology.
24	PO-9.1) Behaving in accordance with ethical principles, awareness of professional and ethical responsibility.
25	PO-9.2) To gain knowledge about the standards used in engineering applications.
26	PO-10.1) Information on business practices such as project management, risk management and change management.
27	PO-10.2) Awareness of entrepreneurship and innovation.
28	PO-10.3) Information on sustainable development.
29	PO-11.1) Universal and social health, environment and safety of engineering applications information about the effects on the problems and the problems reflected in the engineering field of the era.
30	PO-11.2) Awareness of the legal consequences of engineering solutions.

Curriculum

1. Year - Fall Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM1011		Introduction to Computer Science	3	0	2	4	6
FIZ1001		Physics 1	3	0	2	4	6
MAT1071		Mathematics 1	3	2	0	4	6
MAT1320		Linear Algebra	2	0	0	2	4
BLM1991		Occupational Health and Safety 1	2	0	0	2	2
MDB1031		Advanced English 1	3	0	0	3	3
						Total:	27

1. Year - Spring Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM1031		Structured Programming	3	0	2	4	6
FIZ1951		Semiconductor Physics for Engineering	3	0	0	3	6
BLM1022		Numerical Analysis	3	0	0	3	6
MAT1072		Mathematics 2	3	2	0	4	6
BLM1033		Circuit Theory and Electronical Devices	3	0	2	4	6
MDB1032		Advanced English 2	3	0	0	3	3
						Total:	33

2. Year - Fall Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM2012		Object Oriented Programming	3	0	2	4	6
BLM2611		Logic Circuits	3	0	2	4	6
BLM2642		Differential Equations for Computer Engineers	3	0	0	3	6
BLM2011		Statistics and Probability	3	0	0	3	6
BLM2521		Discrete Mathematics	3	0	0	3	6
USS-2G		University Social Elective -1	3	0	0	3	3
						Total:	33

2. Year - Spring Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM2512	☑	Data Structures and Algorithms	3	0	2	4	6
	Önk:	BLM1011 Introduction to Computer Sciences					
BLM2022	☑	Computer Organization	3	0	0	3	4
	Önk:	BLM2611 Logic Circuits					
BLM2041	☑	Signals and Systems for Computer Engineers	3	0	0	3	6
	Önk:	BLM2642					
BLM2042		System Analysis and Design	2	0	0	2	3
BLM2502		Theory of Computation	3	0	0	3	6

BLM1992		Occupational Health and Safety 2	2	0	0	2	2
						Total:	27

3. Year - Fall Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM3011	☑	Operating Systems	2	2	0	3	5
	Önk:	BLM1031					
BLM3041		Database Management	3	0	2	4	6
BLM3021	☑	Algorithm Analysis	2	0	2	3	5
	Önk:	BLM2512					
BLM3061		Microprocessor Systems and Assembly Language	3	0	2	4	5
BLM3042		Seminar and Professional Ethics	3	0	0	3	3
BLM3002		General Internship	0	0	0	0	3
TDB1031		Turkish language 1	2	0	0	0	2
						Total:	29

3. Year - Spring Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM3051		Data Communication and Computer Networks	3	0	0	3	4
BLM3010	☑	Computer Projects	2	2	0	3	4
	Önk:	BLM2042					
MES1-3B		Occupational Elective 1-1	3	0	0	3	8
BLM3722	☑	Software Engineering	3	0	0	3	4
	Önk:	BLM2042 System Analysis and Design					
BLM3510		Artificial Intelligence	3	0	0	3	4
BLM4002		Professional Intership	0	0	0	0	3
SOS1-3B		Social Elective 1-1	3	0	0	3	4
						Total:	31

4. Year - Fall Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
MES2-4G		Occupational Elective 2-1	0	2	0	1	3
UMS-4G		University Professional Elective	3	0	0	3	5
MES1-4G		Occupational Elective 1-2	3	0	0	3	8
MES1-4G		Occupational Elective 1-3	3	0	0	3	8
ATA1031		Principles of Atatürk and History of Modern Turkey 1	2	0	0	0	2
USS-4G		University Social Elective -2	3	0	0	3	3
						Total:	29

4. Year - Spring Semester

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS

BLM9000	☑	Graduation Thesis	0	8	0	4	8
	Önk:	BLM3010 Computer Projects					
MES1-4B		Occupational Elective 1-4	3	0	0	3	8
MES1-4B		Occupational Elective 1-5	3	0	0	3	8
USS-4B		University Social Elective 1-3	3	0	0	3	3
ATA1032		Principles of Atatürk and History of Modern Turkey 2	2	0	0	0	2
TDB1032		Turkish language 2	2	0	0	0	2
						Total:	31
						Program Total ECTS:	240

Occupational Elective 1 Courses

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM3110		Special Topics in Computer Engineering 1	3	0	0	3	8
BLM3630		Special Topics in Computer Engineering 2	3	0	0	3	8
BLM3320		Special Topics in Computer Engineering 3	3	0	0	3	8
BLM3120		Information Retrieval and Web Search Engines	3	0	0	3	8
BLM3130		Introduction To Game Development	3	0	0	3	8
BLM3520		Introduction to Mobile Programming	3	0	0	3	8
BLM3580		System Programming	3	0	0	3	8
BLM3590		Statistical Data Analysis	3	0	0	3	8
BLM3620		Digital Signal Processing	3	0	0	3	8
BLM3720		Introduction to Computer Graphics	3	0	0	3	8
BLM3750		File Organization	3	0	0	3	8
BLM3760		Introduction to Expert Systems	3	0	0	3	8
BLM3780		Database Management system Implementation	3	0	0	3	8
BLM3810		Introduction to Bioinformatics	3	0	0	3	8
BLM4110		Web Services and Service Oriented Architecture	3	0	0	3	8
BLM4120		Big Data Processing and Analytics	3	0	0	3	8
BLM4130		Formal Languages and Automata	3	0	0	3	8
BLM4500		Operating System Architectures for Large Scale Computing Systems	3	0	0	3	8
BLM4520		Introduction to Neural Networks	3	0	0	3	8
BLM4530		Fuzzy Logic	3	0	0	3	8
BLM4540		Image Processing	3	0	0	3	8
BLM4580		Introduction to Natural Language Processing	3	0	0	3	8
BLM4610		Computer Architecture	3	0	0	3	8
BLM4710		Management Information System	3	0	0	3	8
BLM4760		Distributed Systems	3	0	0	3	8

BLM4770		Software Quality and Testing	3	0	0	3	8
BLM4780		Current Topics in Database Systems	3	0	0	3	8
BLM4800		Introduction to Data Mining	3	0	0	3	8
BLM4830		Introduction to Robot Technologies	3	0	0	3	8
BLM4860		Compiler Design	3	0	0	3	8
BLM4900		Advanced Network Programming	3	0	0	3	8
BLM4920		Real Time Systems	3	0	0	3	8
BLM4992		Vocational Education in Operation 1	3	0	0	3	8
BLM4993		Vocational Education in Operation 2	3	0	0	3	8
BLM4994		Vocational Education in Operation 3	3	0	0	3	8
BLM2021		Low Level Programming	3	0	0	3	8
BLM3022		Computer Networking Technologies	3	0	0	3	8
BLM4011		Security of Computer Systems	3	0	0	3	8
BLM4021		Embedded Systems	3	0	0	3	8
BLM4590		Discrete Event Simulation	3	0	0	3	8
BLM3740		Operational Research	3	0	0	3	8
BLM4140	☑	Wireless and Mobile Networks	3	0	0	3	8
	Önk:	BLM2011 Statistics and Probability, BLM3051 Data Communication and Computer Networks					
MTH4700		Software Development with Agile Approaches	3	0	0	3	8
BLM3730		Blockchain Basics	3	0	0	3	8
Occupational Elective 2 Courses							
Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BLM4991		Multidisciplinary Design Project	0	2	0	1	3
EHM4991		Multidisciplinary Design Project	0	2	0	1	3
ELM4991		Multidisciplinary Design Project	0	2	0	1	3
KOM4991		Multidisciplinary Design Project	0	2	0	1	3
BME4991		Multidisciplinary Design Project	0	2	0	1	3
Social Elective 1 Courses							
Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
IKT3322		Macroeconomic Policies	3	0	0	3	5
IKT3562		History of Turkish Administration	3	0	0	3	5
ISL2301		Marketing	3	0	0	3	5
ISL3522		International Marketing	3	0	0	3	5
ISL3230		Operations Research 1	3	0	0	3	5
ISL4611		Business Ethics	3	0	0	3	5
ISL3660		Business Communication	3	0	0	3	5
ISL3930		Corporate Reputation from the Behavioral Perspective	3	0	0	3	5
ISL3040		Team Building and Development in	3	0	0	3	5

		Organizations					
ISL4851		Innovation Management in Management	3	0	0	3	5
ISL4640		Entrepreneurship	3	0	0	3	5
ISL3912		Human Resource Management (Business Administration)	3	0	0	3	5
ISL1611		Introduction to Business	3	0	0	3	5
ISL4420		Sales Management	3	0	0	3	5
ISL3621		Production Management	3	0	0	3	5
ISL4860		Consumer Behaviour	3	0	0	3	5
ISL1622		Behavior Science	3	0	0	3	5
ISL1711		Introduction to Law	3	0	0	3	5
ISL4760		Financial and Cost Accounting	3	0	0	3	5
ISL3631		Career and Work Psychology	3	0	0	3	5

University Social Elective Courses 2

Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
BED3011		Education of Basic Techniques in Basketball	3	0	0	3	3
BED3041		Soccer and Basic Movement Teaching	3	0	0	3	3
BED4031		Principle figures of the folk dances	3	0	0	3	3
BED3051		Education of Basic Techniques in Handball	3	0	0	3	3
BED3012		Education of Basic Techniques in Korfball	3	0	0	3	3
BED4022		Tennis Technic and Tactic Education	3	0	0	3	3
BED3042		Education of Basic Techniques in Volleyball	3	0	0	3	3
BED4032		Education of Fundamental Swimming Techniques	3	0	0	3	3
TRO2261		Turkish Language Teaching Literary Texts	3	0	0	3	3
SNF2112		Geography and geopolitics of Turkey	3	0	0	3	3
ISL2560		Public Relations in Business	3	0	0	3	3
ISL2710		Family Businesses and Institutionalization	3	0	0	3	3
ISL2630		Team Building and Development	3	0	0	3	3
ISL2901		Direct Marketing	3	0	0	3	3
ISL2760		Fundamentals of Logistics Management	3	0	0	3	3
SBP2031		Urban Economics	3	0	0	3	3
ITB2040		Economic Policies and Applications	3	0	0	3	3
ITB3330		Environment and Ecology	3	0	0	3	3
ITB2090		Democracy Culture Principles and Institutions	3	0	0	3	3
ITB3150		History and Cinema	3	0	0	3	3
ITB3020		Introduction to Philosophy	3	0	0	3	3
ITB3040		Political Developments and Social Movements in Twentieth-Century	3	0	0	3	3
ITB3270		Istanbul: Past, Present, and Future	3	0	0	3	3

ILT1611		Techniques Of Photography	3	0	0	3	3
ITB3260		Cultural Studies and Identity	3	0	0	3	3
ITB3420		The Social Structure of Ottoman Empire	3	0	0	3	3
ITB3210		Communication in Contemporary Society	3	0	0	3	3
ITB3220		Modernity and Consumer Society	3	0	0	3	3
ITB3130		Political Ideologies: Theory and History	3	0	0	3	3
ITB2080		Women in Social Transformation	3	0	0	3	3
ISL2170		Accounting Organization	3	0	0	3	3
ITB3010		Sociology	3	0	0	3	3
ITB3550		Human Rights	3	0	0	3	3
ITB3560		Political Philosophy	3	0	0	3	3
ITB3570		Philosophy of Education	3	0	0	3	3
ITB3390		World Civilizations	3	0	0	3	3
ITB2030		Philosophy of Science	3	0	0	3	3
ITB4100		Social Structures and Historical Transformations	3	0	0	3	3
ILT1621		Graphıc Media Design Tools	3	0	0	3	3
SBP2082		Urban Sociology	3	0	0	3	3
SYP2192		Cultural management and Its Agents 2	3	0	0	3	3
SYP3241		Public Relations	3	0	0	3	3
MIM1422		Introduction to History of Art and Architecture	3	0	0	3	3
MIM2421		History of Architecture	3	0	0	3	3
MIM1412		History of Civilization	3	0	0	3	3
HRT2941		History of Geomatic Engineering Science	3	0	0	3	3
ITB2020		History of Science	3	0	0	3	3
INS2462		Traffic Safety	3	0	0	3	3
MDB4011		Introduction to German Language Skills	3	0	0	3	3
MDB4021		German Language Skills	3	0	0	3	3
MAK2100		History of Machine Technology	3	0	0	3	3
ITB3250		Introduction to Psychology	3	0	0	3	3
ITB3360		History of Art	3	0	0	3	3
MTP4760		Dance in Istanbul from the 16th Century to the Present	3	0	0	3	3
GIM4101		Innovation and Entrepreneurship in Engineering	3	0	0	3	3
TDB4011		Effective Communication and Impromptu Presentation Skills	3	0	0	3	3
TDB4031		Oratory and Diction	3	0	0	3	3
TDB4041		Turkish Story and Novel	3	0	0	3	3
ITB1680		Introduction to Polyphonic Music	3	0	0	3	3
TDB4051		Academic Turkish	3	0	0	3	3

DNS1220		Body Awareness and Breathing Techniques	3	0	0	3	3
DNS1240		Yoga and Anatomy	3	0	0	3	3
GIM4151		Innovation and Entrepreneurship	3	0	0	3	3
ITB4040		Volunteering Studies	3	0	0	3	3
TDB4061		Seven Hills İstanbul	3	0	0	3	3
ISL1150		Career Planning	3	0	0	3	3
KIM1052		Chemistry in Life	3	0	0	3	3
CEV3333		Patent and Commecialization	3	0	0	3	3
BED1013		Pilates Basic Training	3	0	0	3	3
MDB1016		Arabic for Beginners 2	3	0	0	3	3
MDB1004		Spanish for Beginners 2	3	0	0	3	3
MKT2201		Personal Mindfullness ans Development	3	0	0	3	3
GRA2024		Sanal Evrene Giriş	3	0	0	3	3
EUT2022		Introduction to NFT	3	0	0	3	3
MDB1001		French for Beginners 1	3	0	0	3	3
MDB1003		Spanish for Beginners 1	3	0	0	3	3
MDB1007		Italian for Beginners 1	3	0	0	3	3
MDB1009		Greek for Beginners 1	3	0	0	3	3
MDB1011		Chinese for Beginners 1	3	0	0	3	3
MDB1013		Japanese for Beginners 1	3	0	0	3	3
SBO1180		Turkish Culture and History	3	0	0	3	3
MDB1015		Arabic for Beginners 1	3	0	0	3	3
MDB1017		Persian for Beginners 1	3	0	0	3	3
OKL2350		Nutrition and Health	3	0	0	3	3
MDB1019		Russian for Beginners 1	3	0	0	3	3
RPD2000		Addiction and Addiction Control	3	0	0	3	3
SBP2020		Earthquake and Planning	3	0	0	3	3
SBO1120		Turkish Cultural Geography	3	0	0	3	3
INS4910		Disaster Information and Awareness	3	0	0	3	3
TRO2730		Media Literacy	3	0	0	3	3
MDB1010		Greek for Beginners 2	3	0	0	3	3
CEV3334		Environment and Human	3	0	0	3	3
BTO1910		Current Practices in Educational Technologies	3	0	0	3	3
MAT4279		Fundamental Rights and Responsibilities in Higher Education	3	0	0	3	3
FBO2260		Sustainability and Education	3	0	0	3	3
MDB1002		French for Beginners 2	3	0	0	3	3
IMO2150		Concepts and Proofs in Linear Algebra	3	0	0	3	3
ING2350		English Academic Writing and Presentation Skills	3	0	0	3	3

MDB1008		Basic Italian 2	3	0	0	3	3
SNF2210		Youth and Education	3	0	0	3	3
SBO1190		Fairy Tales and Storytelling	3	0	0	3	3
SBO1230		Philosophy with Children	3	0	0	3	3
SBO1240		Environmental Citizenship and Education	3	0	0	3	3
TRO2281		Turkish Language History	3	0	0	3	3
EGT1022		Social anthropology	3	0	0	3	3
EGT4041		Education management	3	0	0	3	3
EGT2031		Human Resources Management	3	0	0	3	3
MTM3611		History of Mathematics	3	0	0	3	3
University Occupational Elective Courses 2							
Code	Req.	Title	Lecture	Practical	Laboratory	Local Credit	ECTS
YZM4015		Introduction to Artificial Intelligence	3	0	0	3	5
BYM4721		Nanotechnology in Bioengineering	3	0	0	3	5
SBU3001		Basic Issues in International Relations	3	0	0	3	5
IKT3610		Energy and Natural Resources Economics	3	0	0	3	5
EHM4370		Microprocessor Based System Design	3	0	0	3	5
EHM4220		Satellite Communication	3	0	0	3	5
EHM4270		Cellular Communication Systems 1	3	0	0	3	5
GIM4322		Economics of Energy	3	0	0	3	5
GIM4392		Engineering Economics	3	0	0	3	5
KIM3557		Enviromental Chemistry and Technology	3	0	0	3	5
KMM3561		Technical Communication	3	0	0	3	5
ISL3660		Business Communication	3	0	0	3	5
CEV4501		Natural Treatment	3	0	0	3	5
MAK4482		Industrial Automation	3	0	0	3	5
CEV4111		Environmental and Public Health	3	0	0	3	5
HRT4332		Navigation And Kinematic Positioning	3	0	0	3	5
MIM4341		Space and History in cinema	3	0	0	3	5
ELM4010		Introduction to Smart Grids	3	0	0	3	5
SBP1300		Readings on Cities	3	0	0	3	5
SBP4310		Project Management Process in Participatory City Management	3	0	0	3	5
KVK4412		Cultural Heritage Management	3	0	0	3	5
BME4142		Physiological Control Systems	3	0	0	3	5
IKT3820		Social Policy Economics	3	0	0	3	5
ISL3940		Fundementals of Actuarial Mathematics	3	0	0	3	5
INS3841		Introduction to Construction Legislation	3	0	0	3	5
BLM4400		Contemporary Topics in Comp. Engineering	3	0	0	3	5
BLM1012		Introduction To Procedural Programming	3	0	0	3	5

BME4110		Quantum Physics for Engineers	3	0	0	3	5
TDE3557		Literary Debates in Modern Turkish Literature	3	0	0	3	5
MTM4711		Mathematical Modeling	3	0	0	3	5
ELM4071		Numerical Methods and Applications in Engineering	3	0	0	3	5
KOM4760		Basic Optimization Concepts in Engineering	3	0	0	3	5
KOM4770		Manufacturing Techniques	3	0	0	3	5
GMI3850		Ship-Sourced Marine Pollution	3	0	0	3	5
GMI3860		Structure Dynamics	3	0	0	3	5
IST3557		Statistics and Scientific Thinking	3	0	0	3	5
MAT3557		Encryption	3	0	0	3	5
FIZ3557		Physics in Life	3	0	0	3	5
MBG3557		Evolution and Molecular Ecology	3	0	0	3	5
MEM4131		Material World	3	0	0	3	5
KVK4422		Museology and Museography	3	0	0	3	5
GDM4309		Food Literacy	3	0	0	3	5
MKT4403		Mechatronics System Integration	3	0	0	3	5
END4393		Risk Management	3	0	0	3	5

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Occupational Health and Safety 1	BLM1991	2	2	2	0	0

Prerequisite	None

Semester	Fall

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	
Asistant(s)	

Course Objectives	The main aim of the course is to educate the students about occupational health and safety, to get involved in accidents and occupational diseases within the Department of Electronics and Communication Engineering, to inform about the Law on Occupational Health and Safety No. 6331 and as a result the participation of students in occupational health and safety and the creation of safe culture.
Course Content	Basic Rights and Obligations, Occupational Health and Safety Law No. 6331, Obligations of Employers, Employee Responsibilities, Employee Participation and Informing, OHSAS / TS 18001, Occupational Health and Safety Policy, Occupational Accidents and Occupational Diseases Prevention, Occupational Health and Safety Culture Health and Safety Signs, Health and Safety Education and Communication at Work, National and International Organizations and Contracts, Occupational Health and Safety Committees, Occupational Health and Safety Services, Emergency Management, Risk Management and Evaluation, Health and Safety Management System, Basic First Aid Information, Occupational Health and Safety in Education
Recommended Optional Program Components	None

Course Learning Outcomes

1	Student has knowledge about Occupations and occupational diseases
2	Learn legal rights and responsibilities in occupational health and safety
3	Has knowledge about the Law No. 6331 on Occupational Health and Safety and its applications
4	Know the duties and responsibilities of the occupational safety specialist and the occupational physician, the work of the occupational health and safety body, and the importance of risk assessment
5	Plays an active role in creating a Job Safety Culture by creating safe behaviors and habits, knows what to do in case of an accident or emergency.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation

1	Concepts and Rules of Occupational Health and Safety, Occupational Health and Safety in Turkey and in the World, Historical Development of Occupational Health and Safety in Turkey and in the World. Contemporary business health and safety principles	
2	Occupational health and safety culture, Behavior oriented management, Lifelong learning in occupational health and safety, Integrated approach to occupational health and safety, Occupational health and safety in business management, Risk prevention culture at workplace, Safety culture prominence and place in everyday life, and continuation	
3	Constitution law, statute, regulation, communiqué, circular and directive concepts. Law No. 6331 on Occupational Health and Safety, Employers' obligations, Employee obligations	
4	Definition and general concepts of Occupational Health and Safety Law No. 6331, hazard, risk, risk assessment proactive approach, reactive approach, prevention, work accident, support staff, employee, employee representative	
5	Occupational Health and Safety Services, Occupational Safety and Health Specialist, Occupational Health and Safety, Occupational Health and Safety Unit, Common Health and Safety Unit, Registered Notebook	
6	The duties, powers and responsibilities of contemporary approaches to occupational health and safety, occupational safety specialist, workplace physician, other health personnel, employee representative	
7	Risk prevention principles, Priority ordering in health and safety, Prevention practices in the source, Protection applications for the environment, Protection applications for the person I	
8	Midterm 1 / Practice or Review	
9	Risk prevention principles, Priority ordering in health and safety, Prevention practices in the source, Protection applications for the environment, Protection applications for the person II	
10	TS 18001 Occupational Health and Safety Management System, Management system components, Occupational health and safety policy, PUKO cycle	
11	Health and Safety Signs, Prohibition Signs, Warning Signs, Demand Signs, Emergency and First Aid Signs, Use of Health and Safety Signs, Communication with Health and Safety Signs	
12	Occupational health and safety board, occupational health and safety council	
13	Health Surveillance and Occupational Diseases, Health surveillance concept and its application, Occupational disease concept, Occupational disease types and causes, Occupational diseases protection	
14	Occupational health and safety education and communication, Characteristics and techniques of adult education, Qualifications and period of education, Occupational health and safety and vocational education of employees, Basic concepts about effective communication process in workplace	

15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	1	13
Special Course Internship (Work Placement)			
Homework Assignments	2	5	10
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	8	8
Final (Examination Duration + Examination Prep. Duration)	1	8	8
		Total Workload	67
		Total Workload / 30(h)	2.23
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Occupational Health and Safety 2	BLM1992	2	2	2	0	0

Prerequisite	None

Semester	Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	
Asistant(s)	

Course Objectives	The main purpose of the course is to introduce the risk factors having effects on occupational accidents and diseases, and to teach how to perform the evaluation of risks which have an important role on avoiding such situations. Simultaneously, to get involved in accidents and occupational diseases within the Department of Control and Automation Engineering, to inform about the Law on Occupational Health and Safety No. 6331 and as a result the participation of students in occupational health and safety and the creation of safe culture are also included in the aims of the course.
Course Content	Risk Management, Risk Evaluation, Risk Analysis, Risk Perception, Psychosocial Risk Factors, Physical Risk Factors, Ergonomic Risk Factors, Chemical Risk Factors, Risk Evaluation Methods, Risk Control Steps, Risk Evaluation Stages, Risk Evaluation Documentation, Risk Evaluation Application, Working on Tools with Screen, Ergonomic Work, Protection from Occupational Musculoskeletal diseases
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student knows the concepts of danger, risk, near miss, event / case, accident and the difference between them
2	The student learns why the risk assessment is performed, its function, who and how it is done
3	The student knows the psychosocial, biological, chemical, physical hazards and knows the principles of protection from these hazards
4	The student has an idea and can contribute to a healthier work environment
5	The student detects the hazards earlier and can be more sensitive about the precautions, knows what to do or not in case of an accident or emergency

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation

1	Risk Management and Evaluation, Risk Management and General Management, Hazard and Risk Concepts, Risk Sources and Risks, Risk Assessment as a Part of Risk Management, Basic Principles of Risk Assessment	
2	Risk Assessment, Control, Measurement and Research, Risk Assessment Stages, Risk Control Stages, Risk Assessment Team and Employee Participation, Risk Mitigation Methods	
3	Risk Assessment Methods, Qualitative Methods, Quantitative Methods, Fine-Kinney Method, L Type Matrix, X Type Matrix, Fta, Eta, Cca, Fmea, Jsa, Hazop, Pra, Pha	
4	Occupational Health and Safety in Working with Tools with Screen, Risks and Safeguards in Working with Tools with Screen, Correct Seating, Protection of Eyes, Eyes During Working Shortly, Resting Habit, Interrupted Rests and Exercises	
5	Ergonomics, Lesson Work Ergonomic Work Place Arrangement And Cautions, Office Ergonomics, Cognitive Ergonomics, Physical Ergonomics, Administrative Ergonomics, Anthropometry And Working Environment Design	
6	Psychosocial Risk Factors, Stress, Factors Causing Stress, Stress Stages, Stress Protection Methods, Effects of Work Stress on Health, Long Term Effects of Stress, Psychological Harassment (Mobbing) and Coping Methods	
7	Risk Communication, Risk Perception, Informing Employees, Taking Employee Opinions, Communication and Persuasion I	
8	Midterm 1 / Practice or Review	
9	Risk Communication, Risk Perception, Informing Employees, Taking Employee Opinions, Communication and Persuasion II	
10	Physical Risk Factors, Noise, Vibration, Pressure, Lighting, Thermal Comfort (Working in Humidity, Hot or Cold, Heating and Ventilation), Radiation, Ionized Rays, Non-ionized Rays	
11	Chemical Risk Factors and Protection Methods, Protection Methods, Carcinogenic and Mutagenic Materials, Hazardous Area Classes	
12	Fire and Protection Methods, Safety in Fire, Classification and Extinguishing of Fire, Emergency Management, Preparation of Emergency Plans and Transfer to Employees, Measures and Exercises, First Aid and Emergency Response, Hazard Communication	
13	Occupational Health and Safety at University, Hazards in Classrooms, Hazards in Cafeteria and Cantiles, Psychosocial Hazards Biological Hazards, Chemical Hazards, Electrical Hazards	
14	Risk Assessment Practices	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		

Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	1	13
Special Course Internship (Work Placement)			
Homework Assignments	2	5	10
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	8	8
Final (Examination Duration + Examination Prep. Duration)	1	8	8
		Total Workload	67
		Total Workload / 30(h)	2.23
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Algorithm Analysis	BLM3021	3	5	2	0	2

Prerequisite	BLM2512

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Elif Karslıgil
Instructor(s)	M. Elif Karslıgil, M. Amaç Güvensan
Asistant(s)	

Course Objectives	Course Objectives The goal of this course is to introduce advanced techniques for the design and analysis of major classes of algorithms, and explores a variety of applications.
Course Content	Course Content Fundamentals of the Analysis of Algorithm Efficiency, Asymptotic Notations, Analysis of Divide and Conquer Algorithms, Hashing Algorithms, Graph Algorithms, Dynamic Programming, Backtracking, P, NP and NP-Complete Problems
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to analyze a given algorithm theoretically and practically.
2	Students will learn the concepts of time and space complexity, worst case, average case and best case complexities and asymptotic notations.
3	Students will be able to design efficient algorithms for major engineering problems.
4	Students will learn the design and application areas of commonly used advanced algorithms.
5	Students will learn to report and present comprehensive algorithms.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Fundamentals of the Analysis of Algorithms Efficiency	
2	Asymptotic Analysis	
3	Analysis of Nonrecursive and Recursive Algorithms	
4	Analysis of Divide and Conquer Algorithms	
5	Hashing Algorithms 1	
6	Hashing Algorithms 2	
7	Dynamic Programming 1	

8	Midterm 1 / Practice or Review	
9	Dynamic Programming 2	
10	Graph Algorithms	
11	Graph Algorithms - II	
12	Backtracking	
13	Review and applications	
14	NP, NP-Complete, NP-hard Problems	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	4	16
Presentations/Jury		
Project	1	8
Seminar/Workshop		
Mid-Terms	2	36
Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory	2	6	12
Application	2	2	4
Field Work			
Study Hours Out of Class	1	12	12
Special Course Internship (Work Placement)			
Homework Assignments	4	8	32
Quizzes/Studio Critics			
Project	1	16	16

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	15	30
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	154
		Total Workload / 30(h)	5.13
		ECTS Credit	5

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
German Language Skills	MDB4021	3	3	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Modern Languages
Course Coordinator	Betül Özel
Instructor(s)	Betül Özel, Fitnat Nasırlıel
Asistant(s)	

Course Objectives	Students will be able to develop their knowledge of vocabulary, reading, writing, listening and speaking from A1/1 level to A1/2 level.
Course Content	To express themselves with past simple structures. To practice the structures with intermediate level reading texts.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to read simple reading texts
2	Students will be able to write sentences on their own and express themselves with basic structures.
3	Students will be able to express their own needs in the target language.
4	Students will be able to practice daily language in the target language.
5	Use of Technology: They will have the ability to work independently by doing the internet assignments given to them through technology, and they can access all the materials of the course via Moodle.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to the Course	* introducing yourself * doing related worksheets
2	Unit 8 Wohnen und Leben Describing an apartment Grammar: Demonstrative pronouns Pronunciation of compound words Listening activity	* preparing for the pages between 6-8 * defining new words * reviewing the grammar points * doing related worksheets

3	Unit 8 Wohnen und Leben Colors Talking about the places of objects Grammar: prepositions (auf/in/hinter/neben/unter/vor) Listening activity Workbook (p 76-81) Unit 9 Arbeitsplatze Occupations Grammar: “haben” in past tense	* preparing for the pages between 9 -17 * deciding on the objects needed for professions * bringing those objects in class or drawing pictures of the objects * defining new words * reviewing the grammar points * doing related worksheets
4	Unit 9 Arbeitsplatze Talking about occupations/jobs Grammar: personal pronouns Past simple tense Listening activity Workbook (p 82-87)	* preparing for the pages between 18- 25 * defining new words * reviewing the grammar points * doing related worksheets
5	Unit 10 Einkaufen Food and clothing Shopping dialogues Grammar: Uncountable words Listening activity	* preparing for the pages between 26-31 * bringing a süpermarket brochure in class * defining new words * reviewing the grammar points * doing related worksheets
6	Unit 10 Einkaufen Questionnaire about food and clothing Grammar: Modal verbs indefinite pronouns Listening activity Workbook (p 88-93) Unit 11 Stadt und Verkehr Address description Transportation Grammar: Imperative Listening activity	* preparing for the pages between 32-40 *bringing a city map in class * defining new words * reviewing the grammar points * doing related worksheets
7	Unit 11 Stadt und Verkehr Reading and comprehension questions Grammar: prepositions (nach, aus,bei,mit,seit,von,zu) Listening activity Workbook (p 94-99) Unit 12 Gesundheit Organs and illnesses Grammar: Modals Listening activities	* preparing for the pages between 41- 49 * defining new words * reviewing the grammar points * doing related worksheets
8	Midterm 1 / Practice or Review	* preparing for the pages between 50-55 * defining new words * reviewing the grammar points * doing related worksheets
9	MIDTERM	NA
10	Unit 13 My Neighbours Talking about the apartment and neighbours Rules at home Grammar: Conjunctions Negative structures (nicht und kein) Modals	* preparing for the pages between 56-61 * analyzing the pictures and answering the questions * defining new words * reviewing the grammar points * doing related worksheets
11	Unit 13 My Neighbours Questionairre: Classroom rules Grammar structures for helping each other Listening activity Workbook (p 106-111)	* preparing for the pages between 62-65 * defining new words * reviewing the grammar points * doing related worksheets
12	Unit 14 Reisen in D-A-CH Talking about travels Types of travel Reading text&coprehension questions Workbook (p 100-105)	* preparing for the pages between 66-69 * bringing a souvenir, postcard or photo from a place visited * defining new words * reviewing the grammar points * doing related worksheets
13	Unit 14 Reisen in D-A-CH Writing postcards Buying round trip tickets Planning trips Workbook (112-122) Start Deutsch 1 test	* preparing for the pages between 70-75 * defining new words * reviewing the grammar points * doing related worksheets

14	Student Presentations	* Preparing for the presentation of a given topic and making a presentation
15	Final	* Preparing for the presentation of a given topic and making a presentation
16	FINAL	NA

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	70	
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	15
Presentations/Jury	1	15
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	16	3	48
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	1	14
Special Course Internship (Work Placement)			
Homework Assignments	1	2	2
Quizzes/Studio Critics			
Project			
Presentations / Seminar	1	3	3
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	4	4
Final (Examination Duration + Examination Prep. Duration)	1	5	5

Total Workload	76
Total Workload / 30(h)	2.53
ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to German Language Skills	MDB4011	3	3	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Modern Languages
Course Coordinator	Betül Özel
Instructor(s)	Betül Özel, Fitnat Nasırlıel
Asistant(s)	

Course Objectives	Students will be able to learn grammar and vocabulary knowledge German at beginner level (A1/1)
Course Content	The course content includes learning basic German grammar structures and sentences which students can use in daily life.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Reading: Students will be able to read and understand simple reading texts.
2	Writing: Students will be able to write letters at beginner level, introduce themselves to the others.
3	Listening and Speaking: Students will be able to listen to and understand daily conversations and express themselves
4	Vocabulary: : Students will be able to understand and use vocabulary of daily language at beginner level
5	Use of Technology: They will have the ability to work independently by doing the internet assignments given to them through technology, and they can access all the materials of the course via Moodle.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to the Course	* introducing yourself * doing related worksheets
2	Unit 1: Ich und Du -Introducing yourself -The Alphabet Grammar-Verbs Listening	* preparing for the pages between 6-11 * defining new words * reviewing the grammar points * doing related worksheets

3	Unit 1: Ich und Du -Numbers (1-100) - Inclass dialogues Grammar: Making questions Workbook (p 78-83) Unit 2: Im Deutschkurs -Inclass dialogues -Classroom objects Grammar: Articles, negative structure “kein”, personal pronouns Listening activity	* preparing for the pages between 12-19 * finding the target language equivalents fort he objects available in class * defining new words * reviewing the grammar points * doing related worksheets
4	Unit 2 : Im Deutschkurs Question and Answer structures Preparing vocabulary cards Grammar: Verbs- Questions Listening activity Workbook (p. 84-89)	* preparing for the pages 20-26 * defining new words * reviewing the grammar points * doing related worksheets
5	Unit 3 Das bin ich Personal Information Travelling Occupations Telling where you are Questionnaire: Languages spoken in the course Grammar: Adjectives, verbs (suffix -en, ex: brauchen/haben) Listening activity	* preparing for the pages 27-31 * bringing a document about a person * defining new words * reviewing the grammar points * doing related worksheets
6	Unit 3 Das bin ich Telling where you are Questionnaire: Languages spoken in the course Grammar: Negative structure “nicht”, using verb “sein” Listening activity Workbook (p. 90-95) Unit 4 (Auf dem Markt) Shopping Prices Numbers (100- …) Grammar: Plurals Listening activity	* preparing for the pages 32-39 * defining new words * reviewing the grammar points * doing related worksheets * making a list for the goods we find at oddments market and learn new vocabulary
7	Unit 4 (Auf dem Markt) Statistics Talking to neighbours Grammar: Using personal pronouns Listening activity Workbook (p. 96-101) Unit 5 Meine Familie und Ich Family photos Grammar:possessive pronouns Verb “mögen” Listening activity	* preparing for the pages 40 – 49 * defining new words * reviewing the grammar points * doing related worksheets * bringing a family photo in class and learning new vocabulary about family relationship
8	Midterm 1 / Practice or Review	* preparing for the pages 50-56 * defining new words * reviewing the grammar points * doing related worksheets
9	MIDTERM	NA
10	Unit 6 Viel Zeit im Jahr Seasons and months Weather forecast Vocabulary about weather forecast Using “es” Simple Past Tense Listening activity	* preparing for the pages 57-60 * defining new words * reviewing the grammar points * doing related worksheets * searching the vocabulary for seasons and bringing pictures in class
11	Unit 6 Viel Zeit im Jahr Hours Weather forecast Grammar: Adverbs of time Prepositions “ im/um” Listening activity Workbook (p. 108-113)	* preparing for the pages 61-66 * defining new words * reviewing the grammar points * doing related worksheets
12	Unit 7 Von Früh bis Spat Daily activities Date expressions and working hours Grammar: Adverbs of Time Listening activity	* preparing for the pages 67- 71 * defining new words * reviewing the grammar points * doing related worksheets * taking notes on daily routines

13	Unit 7 Von Früh bis Spat Planning weekend activities Grammar: Prepositions “am/von …bis Listening activity Workbook (p. 108-113)	* preparing for the pages 72 -76 * defining new words * reviewing the grammar points * doing related worksheets * preparing a real estate ad
14	Student Presentations	* Preparing for the presentation of a given topic and making a presentation
15	Final	* Preparing for the presentation of a given topic and making a presentation
16	FINAL	NA

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	70	
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	15
Presentations/Jury	1	15
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	16	3	48
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	1	14
Special Course Internship (Work Placement)			
Homework Assignments	1	2	2
Quizzes/Studio Critics			

Project			
Presentations / Seminar	1	4	4
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	4	4
Final (Examination Duration + Examination Prep. Duration)	1	5	5
Total Workload			77
Total Workload / 30(h)			2.57
ECTS Credit			3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Principles of Atatürk and History of Modern Turkey 1	ATA1031	0	2	2	0	0

Prerequisite	None

Semester	Fall

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Distance Learning

Owner Academic Unit	Department of Principles of Atatürk and History of Modern Turkey
Course Coordinator	Mehmet Beşikçi
Instructor(s)	Eray Yılmaz, Gülsema Lüyer, Neslihan Erkan, Şebnem Gelmedi, Zafer Doğan
Asistant(s)	

Course Objectives	To provide students with some examples of a multi-layered point in order to make them able to approach historical events in a multi-dimensional way. To introduce to students certain basic theoretical concepts, discussions and methods of thought of different social sciences, with a particular emphasis on history.
Course Content	Basic political, economic, social and cultural facts of the historical period beginning from the 19th century of the Ottoman Empire and ending by the signing of Lausanne Treaty in 1923 - the fundamental academic interpretations on them.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The students will learn meaning and benefits of historical researches.
2	The students will learn the pre-modern Ottoman history in general.
3	The students will be able to evaluate Ottoman history within the European modernization process.
4	The students will be able to evaluate 19.th century Ottoman history within the context of modernization efforts.
5	Finally, the students will understand and evaluate today in relation to the history of Ottoman Empire and modern Turkey.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction: The Possibilities and the limitations of history: basic concepts	Georg Iggers, Yirminci Yüzyılda Tarihyazımı, s. 1-21
2	Social and Administrative Structure of the Ottoman State, before the attempts of modernization: From 16th to the 18th Century	Erik Jan Zürcher, Modernleşen Türkiye'nin Tarihi, s. 23-38
3	Transformation in the Social and Administrative Structure of the Ottoman State, before the attempts of modernization: 18th Century	Niyazi Berkes, Türkiye'de Çağdaşlaşma, s. 65-80

4	The meaning of the modernization and the formation of the modern state / The beginnings of the attempts of modernization in the Ottoman State: the beginning of the long 19th century	Peter Burke, Tarih ve Toplumsal Kuram, s. 129-137 / Eric Jan Zürcher, Modernleşen Türkiye'nin Tarihi, s. 39-77
5	The Tanzimat Era (1839-1876): The Reconstruction of the centralized state	Şerif Mardin, Türkiye'de Toplum ve Siyaset, s. 288-310
6	The Era of Abdülhamid II (1876-1908): Defensive Modernization	Erik Jan Zürcher, Modernleşen Türkiye'nin Tarihi, s. 109-136
7	The Era of Second Constitutional Monarchy : A Constitutional Revolution	Erik Jan Zürcher, Modernleşen Türkiye'nin Tarihi , s. 139-186
8	Midterm 1 / Practice or Review	
9	The Era of Second Constitutional Monarchy: Pluralism in the Public Sphere	Erik Jan Zürcher, Modernleşen Türkiye'nin Tarihi, s. 186-193
10	The First World War: “Total War” and the rise of the nationalism	Gökçen-Faruk Alpkaya, 20. Yüzyıl Dünya ve Türkiye Tarihi, s. 71-79
11	The General Social and Political Situation in the world and in the Ottoman State after the First World War	Feroz Ahmad, Bir Kimlik Peşinde Türkiye, s.93-98.
12	The War of Independence I: The Political Developments	Eray Yılmaz, İmparatorluktan Ulus Devlete Türkiye Tarihi, s. 189-210
13	The War of Independence II: The Military Developments	Erik Jan Zürcher, Modernleşen Türkiye'nin Tarihi, s. 222-228
14	The Formation and the Contents of the Lausanne Treaty	Eray Yılmaz, İmparatorluktan Ulus Devlete Türkiye Tarihi, s. 215-220
15	Final	
16	Final	NA

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	30
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table			
Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	1	6	6
Quizzes/Studio Critics			
Project			
Presentations / Seminar			0
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	6	6
Final (Examination Duration + Examination Prep. Duration)	1	6	6
		Total Workload	74
		Total Workload / 30(h)	2.47
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Principles of Atatürk and History of Modern Turkey 2	ATA1032	0	2	2	0	0

Prerequisite	None

Semester	Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Distance Learning

Owner Academic Unit	Department of Principles of Atatürk and History of Modern Turkey
Course Coordinator	Mehmet Beşikçi
Instructor(s)	Eray Yılmaz, Gülsema Lüyer, Neslihan Erkan, Şebnem Gelmedi, Zafer Doğan
Asistant(s)	

Course Objectives	To inform students about political, economic, social and cultural facts of the historical period beginning from 1923 to the present. To provide students with some significant examples of a multi-layered point of view in evaluating historical events. With an interdisciplinary perspective, to introduce to students some basic theoretical concepts, discussions and methods of thought of different social sciences, with particular emphasis on history.
Course Content	Basic political, economic, social and cultural facts of the historical period beginning from 1923 to the present; fundamental academic interpretations on them.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The students will acquire a perspective to evaluate the 20.th century.
2	The students will evaluate the political, economic and cultural policies of the early republican era.
3	The students will evaluate the political, economic and cultural policies of the Democratic Party era.
4	The students will evaluate the political, economic and cultural policies after the military coup of 1980.
5	The students will evaluate today within the context of Republican history.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	An overview of the 20th century	Eric Hobsbawm, Kısa 20 Yüzyıl: Aşırılıklar Çağı, s. 13-31
2	Political Life between 1923-1945	Çağdaş Türkiye 1908-1980, s. 85-154
3	The formation of the Republican Ideology and the Kemalist Principles	Cumhuriyet Dönemi Türkiye Ansiklopedisi, c.1 s. 86-88
4	The Social and Cultural Transformation between 1923-1950	Modern Türkiye'de Siyasi Düşünce: Modernleşme ve Batıcılık, s.382-402

5	The Turkish Economy between 1923-1945	Cumhuriyet Dönemi Türkiye Ansiklopedisi, c.2 s. 412-418
6	International Relations of Turkey between 1923-1945	Erik Jan Zürcher, Modernleşen Türkiye’nin Tarihi, s. 291-298
7	The Passage of Turkey to the plural political system: 1945-1950	Eray Yılmaz, İmparatorluktan Ulus Devlete Türkiye Tarihi, s. 265-270
8	Midterm 1 / Practice or Review	
9	1950-1960: Political Developments During the Years of Democratic Party	Erik Jan Zürcher, Modernleşen Türkiye’nin Tarihi, s. 321-350
10	Politics in Turkey between 1960-1980	Feroz Ahmad, Bir Kimlik Peşinde Türkiye, s.147-182.
11	Economic Development and Social change in Turkey between 1960-1980	Cumhuriyet Dönemi Türkiye Ansiklopedisi, c.4 s. 1065-1073
12	The Military Intervention in 1980 and the Rise of the Neo-Liberalism	Feroz Ahmad, Bir Kimlik Peşinde Türkiye, s. 185-198
13	Gender Politics in Turkey	Yüzyıl Biterken Cumhuriyet Dönemi Türkiye Ansiklopedisi, c.13 s. 757-764
14	The Constitutions in Turkey	Cumhuriyet Dönemi Türkiye Ansiklopedisi, c.1 s. 18-54
15	Final	
16	Final	NA

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	30
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	1	6	6
Quizzes/Studio Critics			
Project			
Presentations / Seminar			0
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	6	6
Final (Examination Duration + Examination Prep. Duration)	1	6	6
		Total Workload	74
		Total Workload / 30(h)	2.47
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Discrete Mathematics	BLM2521	3	6	3	0	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Banu Diri
Instructor(s)	Banu Diri, Ahmet Elbir
Asistant(s)	

Course Objectives	To learn a particular set of mathematical facts and how to apply them and how to think mathematically.
Course Content	Logic; Sets and Functions; Fundamentals of Algorithms; Integers and matrices; Counting Techniques; Chromatics Polinomials; Graphs; Trees; Boolean Algebra; Finite-State Machine with/without Output
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will learn the basics of creating a mathematical model.
2	The student will learn mathematical concepts and terminology.
3	The student will know how to analyze recursive definitions, and how to use it.
4	The student will understand how to use different types of discrete structures.
5	The student will know how to perform mathematical proofs.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	The Language of Mathematics	Discrete Mathematics and Its Applications Section 1
2	Logic and Sets	Discrete Mathematics and Its Applications Section 1
3	Functions	Discrete Mathematics and Its Applications Section 2
4	Algorithms and Complexity of Algorithms	Discrete Mathematics and Its Applications Section 2
5	Mathematical Reasoning	Discrete Mathematics and Its Applications Section 3

6	Counting	Discrete Mathematics and Its Applications Section 4
7	Advanced Counting Techniques	Discrete Mathematics and Its Applications Section 5
8	Midterm 1 / Practice or Review	midterm
9	Relations	Discrete Mathematics and Its Applications Section 6
10	Choramatics Polinomials	Discrete Mathematics, R. Johnsonbaugh
11	Trees and their Applications-1	Discrete Mathematics and Its Applications Section 8
12	Trees and their Applications-2	Discrete Mathematics and Its Applications Section 8
13	Graph Theory-1	Discrete Mathematics and Its Applications Section 7
14	Graph Theory-2	Discrete Mathematics and Its Applications Section 7
15	Final	Discrete Mathematics and Its Applications Section 9
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	60
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42

Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	3	42
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	30	60
Final (Examination Duration + Examination Prep. Duration)	1	30	30
		Total Workload	174
		Total Workload / 30(h)	5.80
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Formal Languages and Automata	BLM4130	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Not Assigned
Instructor(s)	
Asistant(s)	

Course Objectives	To improve programming language skills by achieving basic knowledge of classification and definition of languages, and relation to automata and their functions.
Course Content	Alphabet, Language, Grammar, Classification of Grammars, Chomsky Hiearchy, Regular Grammars, Context Free Grammars, CFG and BNF, Parse Tree, Left Recursion and Elimination, Pumping Dilemma, Decision Problem, Normal Forms, Pushdown Automata, Context Sensitive Grammars, Linear Bounded Automata, Unrestricted Grammars, Turing Machine, Curch Turing Hipotesis, Codes, Schutzenberger Criteria, Sardinas Patterson Algorithm, Prefix Codes, Bounded Delay Codes, Optimal Codes And Huffman Algorithm.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students know the hierarchy of formal grammars.
2	Students will earn motivations to learn and design new programming languages.
3	Students will haveconfidence in fundamental subjects such as finite automata and Turing machines.
4	Students will learn the limitations of the codes generated by grammars
5	Students will be able to differantiate the grammars that can be used as programming languages

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Course Introduction and Basic Terms	
2	Grammars And Chomsky Hierarchy	Harrison, M.A
3	Regular Grammars	Hopcroft
4	Context Free Grammars, Parse Trees	Hopcroft
5	CFG Notation	Hopcroft

6	BNF Notation	Harrison, M.A
7	Left Recursion and Elimination, Pumping Dilemma	Hopcroft
8	Midterm 1 / Practice or Review	Hopcroft
9	Midterm	Course Notes
10	Context Sensitive Grammars, Linear Bounded Automada	Hopcroft
11	Unrestricted Grammars, Turing Machines	Hopcroft
12	Turing Machines, Church-Turing Hypothesis	Hopcroft
13	Codes, Schutzenberger Criteria	Harrison, M.A
14	Prefix Codes, Bounded Delay Codes	Harrison, M.A
15	Final	Course Notes
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	5	25
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	5	10	50

Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	35	35
		Total Workload	239
		Total Workload / 30(h)	7.97
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Computer Science	BLM1011	4	6	3	0	2

Prerequisite	None

Semester	Fall

Course Language	Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Amaç Güvensan
Instructor(s)	M. Amaç Güvensan, Göksel Biricik, H.İrem Türkmen
Asistant(s)	Sultan Sevgi TURGUT , Meliha Gizem ÇELİK, Rukiye BAŞKARA, Muzaffer Kaan YÜCE

Course Objectives	To teach the basics of Computer Engineering and the algorithm concepts to zero-experienced students.
Course Content	History of Computers ; The Basics of Computer Science and Computer Engineering ; Software and Hardware Comcepts ; Computer Architecture ; Data Manipulation ; Signed and Unsigned Integers ; Floating Point Numbers ; Number Systems ; Introduction to Algorithms ; Flowcharts ; Pseudo Code; Input/Outpput ; Arithmetic Operations ; Controls ; Loops ; Introduction to Coding ; Data Types ; Arrays ; Min-Max Problem ; Strings ; Multi-dimensional Arrays ; Search Algorithms ; Complexity of Algorithms
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will understand fundamental underlying principles concepts of computer engineering.
2	Students will understand how to design correct and efficient algorithms.
3	Students will learn about the process of writing and debugging a program.
4	Students will learn how to describe the devised algorithms as flowcharts.
5	Students will be able to know different branches of computer engineering.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	History of Computers / Fundamental Concepts of Computer Sciences and Engineering	
2	Computer Architecture / Number Systems and Conversion	
3	Data Storage and Manipulation / Signed and Unsigned Numbers / Integers and Floating Point Numbers	
4	Algorithm Concept / Flow Charts / Pseudo Code	

5	Input/Output / Arithmetic Operations / Control Flows	
6	Loops	
7	Introduction to Coding / Basic Data Types	
8	Midterm 1 / Practice or Review	
9	Arrays	
10	Strings	
11	Multi-dimensional Arrays	
12	Practice for Coding Basic Algorithms	
13	Search Algorithms	
14	Sorting Algorithms / Complexity of Algorithms	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	2	8
Homework Assignments	3	12
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	2	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory	10	2	20
Application	3	2	6
Field Work			
Study Hours Out of Class	12	2	24
Special Course Internship (Work Placement)			

Homework Assignments	3	8	24
Quizzes/Studio Critics	2	2	4
Project	1	32	32
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	8	16
Final (Examination Duration + Examination Prep. Duration)	1	16	16
		Total Workload	184
		Total Workload / 30(h)	6.13
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Computer Graphics	BLM3720	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oğuz Altun
Instructor(s)	Songül Varlı
Asistant(s)	

Course Objectives	To teach the computer graphics math and basic information
Course Content	GPU for graphics, shaders and shading language, animation of objects, 3D affine transformations, koordinate transformations, perspective and orthogonal projections, texture mapping.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will understand the fundamentals of computer graphics.
2	The student will gain the ability to perform basic 2D and 3D transformations.
3	The student will be able to do coordinate transformations.
4	The student will learn the infrastructure of mathematics used in computer graphic.
5	The student will gain the ability to realize 3-D projects.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Computer Graphics	Angel and Shreiner Chapter 1
2	Math Review	
3	Introduction to OpenGL	Angel and Shreiner Chapter 2
4	GLSL and Shaders	Angel and Shreiner Chapter 2
5	Input and Interaction	Angel and Shreiner Chapter 3
6	Geometric Objects and Homogeneous Coordinates	Angel and Shreiner Chapter 4
7	Transformations	Angel and Shreiner Chapter 4
8	Midterm 1 / Practice or Review	Angel and Shreiner Chapter 5
9	Transformations	Angel and Shreiner Chapter 4
10	Projection Matrices	Angel and Shreiner Chapter 5

11	Lighting and Shading	Angel and Shreiner Chapter 6
12	Lighting and Shading	Angel and Shreiner Chapter 6
13	Project Presentations	
14	Project Presentations	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	30
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	6	78
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	35	35
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	40	40
Final (Examination Duration + Examination Prep. Duration)	1	48	48

Total Workload	243
Total Workload / 30(h)	8.10
ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Signals and Systems for Computer Engineers	BLM2041	3	6	3	0	0

Prerequisite	BLM2642

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Ali Can Karaca
Instructor(s)	Ali Can Karaca, Ahmet Elbir
Asistant(s)	

Course Objectives	Course Objectives This course is intended to provide fundamental information about signals and systems in both continuous and discrete domain for computer engineers.
Course Content	Course Content Introduction to Fundamental Concepts; Signals and Systems; Linear and Time- Invariant Systems; Time Domain Analysis of Continuous Time Systems; Time Domain Analysis of Discrete Time Systems; Continuous Time Fourier Series; Discrete Fourier Transform; Continuous Time Fourier Transform; Fourier Transform of Some Functions; Sampling; Laplace Transform; Z-Transform.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students understand the basics of signals and systems properties.
2	Students know the continuous and discrete domain signal processing concepts.
3	Students learn how to extract the frequency content of continuous and discrete domain signals.
4	Students know how to analyze linear time invariant systems’ transient and steady state responses.
5	Students learn how to design continuous and discrete time linear time invariant systems.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Definitions of signals and systems, analog and digital systems, analog to digital conversion, sampling concepts, quantization, basic units used in this course.	McClellan ch.1 and 2 Kamen ch. 1
2	Properties and applications of signals, time and frequency domain, classications of signals, signal plotting in programs.	McClellan ch.3 Kamen ch.3

3	Discrete and continuos time signals, types of systems, properties and applications of systems, operations on signals, periodic and aperiodic signals, energy and power signals, basic signals and their properties, complex exponentials, Review of Sine and Cosine Functions. Sinusoidal Signals. Sampling and Plotting Sinusoids.	McClellan ch.11 Kamen ch.3
4	Block diagram representation, some types of systems, interconnection of systems, linearity, homogenity, memoryless, causality, time-invariance, stability, invertibility of systems, differantial and difference systems.	McClellan ch.11 Kamen ch.3
5	Impulse response, LTI systems, convolution integral, properties of convolutions, graphical convolution in continuous systems,	McClellan ch.4 Kamen ch.5
6	Convolution sum, graphical convolution in discrete systems, problem solutions	Kamen ch.5
7	Convolution sum, graphical convolution in continuous systems, problem solutions	Kamen ch.5
8	Midterm 1 / Practice or Review	Kamen ch.6
9	Establishing a General Input-Output Relationship, Obtaining the differential/difference equations for the input-output relations of systems. Solution of differential and discrete equations in the time domain.	Kamen ch.5
10	The Spectrum of a Sum of Sinusoids. Beat Notes. Periodic Waveforms. Fourier Series Analysis and Synthesis. Time-Frequency Spectrum. Frequency Modulation.	McClellan ch.5
11	Continuous-Time Aperiodic Signals. Continuous-Time Fourier Transform. Properties of Continuous-Time Fourier Transform. Discrete-Time Fourier Transform Properties of Discrete-Time Fourier Transform.	McClellan ch.6
12	Sampling. Spectrum View of Sampling and Reconstruction. Discrete-to-Continuous Conversion. The Sampling Theorem.	McClellan ch.7
13	Laplace Transform. Common Laplace Transforms. Properties Of the Laplace Transform. Inverse Laplace Transform. Poles and Zeros in the s-plane.	
14	Z Transform. Common Z Transforms. Properties Of the Z Transform. Inverse Z Transform. Poles and Zeros in the z-plane.	McClellan ch.8
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	3	30

Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	30
Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	3	15	45
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	20	40
Final (Examination Duration + Examination Prep. Duration)	1	25	25
Total Workload			180
Total Workload / 30(h)			6.00
ECTS Credit			6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Special Topics in Computer Engineering 1	BLM3110	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	Hamza Osman İlhan
Asistant(s)	

Course Objectives	This course aims to teach graduate students the special topics in Computer Engineering that are thought to be important in near future.
Course Content	A Special Topic in the field of Computer Engineering is selected and relevant scientific research is done. Within the scope of the course content the followings will be covered in the field of Computer Engineering; the research process (problem identification, data collection, data analysis and interpretation of results) to be examined; major scientific research methods review (experimental method, descriptive method, historical method and the like); conducting literature survey on the identified topics; data collection and evaluation; report writing techniques.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Special topics in Computer Engineering and related concepts will be learned.
2	Special topics in Computer Engineering and related skills for developing applications will be acquired.
3	Special topics in Computer Engineering will be analyzed and the skills to evaluate the applications (performance, scalability, usability criterias) will be gained.
4	This course will provide the ability to comprehend and apply the proposed approaches on the most current topics in Computer Engineering.
5	This course will provide the ability to apply the proposed models on the most current topics in Computer Engineering.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Course	
2	How to conduct literature survey in a special topic of the Computer Engineering	
3	How to set the purpose, assumptions, limitations and definitions in accordance with the special topic of the Computer Engineering - I	

4	How to set the purpose, assumptions, limitations and definitions in accordance with the special topic of the Computer Engineering - II	
5	How to do statistical procedures and analysis on the data of the special topic of the Computer Engineering - I	
6	How to do statistical procedures and analysis on the data of the special topic of the Computer Engineering - II	
7	How to explain features related with the interpretations of the findings in the special topic of the Computer Engineering - I	
8	Midterm 1 / Practice or Review	
9	How to explain features related with the interpretations of the findings in the special topic of the Computer Engineering - II	
10	How to do interpretation of the findings in special topic of the Computer Engineering - I	
11	How to do interpretation of the findings in special topic of the Computer Engineering - II	
12	How to discuss suggestions based on the results of the research in special topic of the Computer Engineering	
13	How to summarize the results of the research in special topic of the Computer Engineering	
14	How to do technical writing in special topic of the Computer Engineering - I	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation	15	5
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	3	15
Homework Assignments		
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	6	84
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics	3	5	15
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	40	40
		Total Workload	241
		Total Workload / 30(h)	8.03
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Computer Architecture	BLM4610	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Songül Varlı
Instructor(s)	Songül Varlı
Asistant(s)	

Course Objectives	Objective in this course is to cover properties of various computer archıtectures and give new techniques in computer architecture to improve system performance.
Course Content	In this course, we cover computer components, pipelining, cache techniques and various properties of computer architectures and teach paralel programming applications.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students can describe fundemental principles of computer architectures
2	Students can differenciate the architectural properties that affect the system performance
3	Students can follow the new techniques and trends to improve the performance
4	Students can analyze the performance of the multiprocessor systems
5	Students can evaluate the system perfomance and experience paralel programming wiith (MPI).

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction; Computer components and functions	
2	Computer Evolution	
3	Cache memory design and mapping algorithms	
4	Internal and External memory	
5	Input/Output	
6	Instruction Set	
7	Addressing modes and formats	
8	Midterm 1 / Practice or Review	
9	Midterm Exam	

10	Pipeline design	
11	RISC Processors and CISC	
12	Multiprocessor Systems	
13	Superscalar processors	
14	Message Passing Interface-MPI	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	4	10
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	2	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	13	3	39
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	6	78
Special Course Internship (Work Placement)			
Homework Assignments	3	15	45
Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	15	30

Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	242
		Total Workload / 30(h)	8.07
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Computer Organization	BLM2022	3	4	3	0	0

Prerequisite	BLM2611 Logic Circuits

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Erkan Uslu
Instructor(s)	Erkan Uslu, Ali Can Karaca
Asistant(s)	

Course Objectives	This course is intended to overview of combinational and sequential circuits and to give information about the technology of computer hardware like memory systems, pipeline processing, cache memories and virtual memory.
Course Content	This course gives information about Central Processing Unit (CPU) and Control Unit design, Input/Output Interface and Communication, memory systems, and management.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students understand the basic design of central processing unit(CPU) and control unit
2	Students can design a communication circuitry between CPU and I/O devices.
3	Students understand the memory systems, cache memories and mapping methods.
4	Students learn how to design a digital circuit.
5	Students learn fundamentals of logic design and their relation to computer organization.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction, Digital Abstraction	Harris Ch1
2	Combinational Logic Design	Harris Ch2
3	Sequential Logic Design	Harris Ch3
4	Hardware Description Language, Practical 1	Harris Ch4
5	Digital Building Blocks, Homework 1	Harris Ch5
6	Instruction Set Architecture	Harris Ch6
7	Microarchitecture I (Single Cycle Processor), Homework 2	Harris Ch7
8	Midterm 1 / Practice or Review	

9	Microarchitecture II (Single Cycle Processor)	Harris Ch7
10	Microarchitecture III (Multi Cycle Processor), Homework 3	Harris Ch7
11	Microarchitecture III (Pipelined Processor), Practical 2	Harris Ch7
12	Microarchitecture IV (Advanced Microarchitecture)	Harris Ch7
13	Memory Systems I (Cache), Homework 4	Harris Ch8
14	Memory Systems II (Virtual Memory)	Harris Ch8
15	Final	Mano Ch13
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	5	30
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			0
Field Work			
Study Hours Out of Class	16	2	32
Special Course Internship (Work Placement)			
Homework Assignments	5	4	20
Quizzes/Studio Critics			0
Project			
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	10	10
		Total Workload	114
		Total Workload / 30(h)	3.80
		ECTS Credit	4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Computer Projects	BLM3010	3	4	2	2	0

Prerequisite	BLM2042

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Ahmet Elbir
Instructor(s)	Sırma Yavuz, Oya Kalipsiz, M. Utku Kalay, Banu Diri, Gökhan Bilgin, M. Fatih Amasyalı, M. Elif Karslıgil, Oğuz Altun, Songül Varlı, Yunus Emre Selçuk, Göksel Biricik, M. Amaç Güvensan, Mehmet Sıddık Aktaş, Erkan Uslu, Ahmet Elbir, Hamza Osman İlhan, H.İrem Türkmen, Furkan Çakmak, Ali Can Karaca, Ayşe Betül Oktay
Asistant(s)	

Course Objectives	Make students to gain experience about hardware and software related subjects with projects that combine them.
Course Content	Ability to solve problems in public and global domains with the help of studied software and hardware topics.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will learn the requirements of computer engineering, analytical thinking and approaching problems to produce algorithmic solutions.
2	Students will able to implement systems development life cycle in the projects.
3	Students will able to revise and improve systems designed by using the results obtained by implementing experiments and solutions.
4	Students will learn effectively writing of their team projects in reports.
5	Students will understand independent learning of new technologies and concepts in order to complete the project.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Project topic research	Yes
2	Determination of the project topic	Yes
3	Evaluation of related works about the project topic	Yes
4	Preparation of the feasibility report	Yes
5	Determination of application details and modules	Yes

6	Database design	Yes
7	Implementation	Yes
8	Midterm 1 / Practice or Review	Yes
9	First project consultation	Yes
10	Implementation	Yes
11	Implementation	Yes
12	Submission of the second progress report	Yes
13	Second project consultation	Yes
14	Implementation	Yes
15	Final	Yes
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	60
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory			
Application	14	2	28
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments			

Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	16	16
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	120
		Total Workload / 30(h)	4.00
		ECTS Credit	4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Information Retrieval and Web Search Engines	BLM3120	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Mehmet Sıddık Aktaş
Instructor(s)	Mehmet Sıddık Aktaş
Asistant(s)	

Course Objectives	Course Objectives The purpose of the Web Data Mining course is to teach students how to mine information from the Web content, Web Structure and Web Usage data. This course also teaches how to use this information to better understand the data.
Course Content	Introduction to Web mining concepts, Internet and the Web graph, Information retrieval and Web search, Link analysis, Web crawling, Web usage mining, Clustering approaches for Web mining, Classification approaches for Web mining.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Web mining concepts will be learned.
2	Web mining skills will be acquired to extract knowledge from Web content, Web Structure and Web Usage data.
3	Web data will be analyzed and prepared for mining and outlier data will be discovered.
4	Web mining techniques will gain reliable information processing skills.
5	Gain the ability to design web search engines.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Course	
2	Association Rule Mining on Web Data	
3	Supervised Learning on Web Data	
4	Unsupervised Learning on Web Data	
5	Partially Supervised Learning on Web Data	
6	Information Retrieval and Web Search Engines	
7	Link Analysis on Web Data	

8	Midterm 1 / Practice or Review	
9	Web Crawling Algorithms	
10	Web Crawling Algorithms	
11	Structured Data Extraction	
12	Opinion Mining	
13	Web Usage Mining	
14	Student Presentations	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	6	15
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	6	84
Special Course Internship (Work Placement)			
Homework Assignments	6	5	30
Quizzes/Studio Critics			
Project	1	50	50

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	13	13
Final (Examination Duration + Examination Prep. Duration)	1	13	13
		Total Workload	232
		Total Workload / 30(h)	7.73
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Graduation Thesis	BLM9000	4	8	0	8	0

Prerequisite	BLM3010 Computer Projects

Semester	Spring

Course Language	English
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	Nizamettin Aydın, Oya Kalipsiz, Banu Diri, M. Elif Karslıgil, Songül Varlı, Sırma Yavuz, Yunus Emre Selçuk, Gökhan Bilgin, Oğuz Altun, M. Fatih Amasyalı, M. Utku Kalay, Mehmet Sıddık Aktaş, M. Amaç Güvensan, Erkan Uslu, Göksel Biricik, Ziya Cihan Tayşi, Hasan Hüseyin Balık, H.İrem Türkmen
Asistant(s)	

Course Objectives	Make students to gain experience about hardware and software related subjects with projects that combine them.
Course Content	Ability to solve problems in public and global domains with the help of studied software and hardware topics.
Recommended Optional Program Components	None

Course Learning Outcomes

1	To fulfill the requirements of computer engineering, analytical thinking and approaching problems to produce algorithmic solutions.
2	Ability to implement systems development life cycle in the projects.
3	Ability to revise and improve systems designed by using the results obtained by implementing experiments and solutions.
4	Students will learn effectively writing of their team projects in reports.
5	Students will understand independent learning of new technologies and concepts in order to complete the project.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Project topic research	Yes
2	Determination of the project topic	Yes
3	Evaluation of related works about the project topic	Yes
4	Preparation of the feasibility report	Yes
5	Determination of application details and modules	Yes

6	Database design	Yes
7	Implementation	Yes
8	Midterm 1 / Practice or Review	Yes
9	First project consultation	Yes
10	Implementation	Yes
11	Implementation	Yes
12	Submission of the second progress report	Yes
13	Second project consultation	Yes
14	Implementation	Yes
15	Final	Yes
16	Presentation of project	Yes

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	60
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours			0
Laboratory			
Application	3	40	120
Field Work			
Study Hours Out of Class	10	13	130
Special Course Internship (Work Placement)			
Homework Assignments			

Quizzes/Studio Critics			
Project			
Presentations / Seminar	2	1	2
Mid-Terms (Examination Duration + Examination Prep. Duration)			0
Final (Examination Duration + Examination Prep. Duration)	2	1	2
		Total Workload	254
		Total Workload / 30(h)	8.47
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Bioinformatics	BLM3810	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Nizamettin Aydın
Instructor(s)	Nizamettin Aydın
Asistant(s)	

Course Objectives	To convey the importance of bioinformatic approaches in solving problems in modern biological research and provide hands on experience using software, critically evaluating results, and interpreting their biological significance.
Course Content	Bioinformatics is a rapidly growing field that integrates molecular biology, biophysics,statistics, and computer science. The course provdes broad overview of bioinformatics with a significant problem-solving component, including hands-on practice using computational tools to solve a variety of biological problems. Topics include: database searching, sequence alignment, gene prediction, RNA and protein structure prediction, construction of phylogenetic trees, comparative and functional genomics
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will understand the fundamental biological questions.
2	Students will understand how to apply appropriate computational tools.
3	Students will understand the basic principles underlying the computational approaches.
4	Students will understand basic principles underlying statistical and mathematical approaches.
5	Students will understand the limitations of available tools and how to critically interpret results.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction. Definition of Bioinformatics. Importance of Bioinformatics.	
2	Overview of Molecular Biology. Cells, Chromosomes, DNA, RNA, Amino Acids, Proteins, Genome, Transcriptome, Proteome.	
3	Perl as a Software Tool, Installing, Debugging, Programming.	
4	Pairwise Sequence Alignment. Relation of Sequences.	

5	Alignment methods (Visual, Brute Force, Dynamic Programming, Word-Based).	
6	Dot plots, Global Alignment, Local Alignment.	
7	Scoring Matrices, Significance of Alignments..	
8	Midterm 1 / Practice or Review	
9	Web Technologies and Perl in Bioinformatics.	
10	Multiple Sequence Alignment. Global Multiple Alignment, progressive global alignment, Iterative methods,	
11	Alignments based on locally conserved patterns.	
12	Local Multiple Alignment, Profile Analysis, Block Analysis, Pattern-searching or statistical methods.	
13	Phylogenetics	
14	Sequence Data File Formats. Data Visualization	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		5
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	4	5
Homework Assignments	3	15
Presentations/Jury	1	5
Project	1	10
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	13	3	39
Laboratory			
Application			
Field Work			

Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	3	15	45
Quizzes/Studio Critics	4	6	24
Project	1	20	20
Presentations / Seminar	1	20	20
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	240
		Total Workload / 30(h)	8.00
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Fuzzy Logic	BLM4530	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Sırma Yavuz
Instructor(s)	Sırma Yavuz
Asistant(s)	

Course Objectives	The aim of the lecture is to teach usage of fuzzy logic in different areas like artificial intelligence techniques and systems.
Course Content	Fuzzy Sets; The Operation of Fuzzy Sets; Fuzzy Relation and Composition; Fuzzy Graph and Relation; Fuzzy Number; Fuzzy Function; Probability and Uncertainty; Fuzzy Logic; Fuzzy Inference; Fuzzy Control; Fuzzy Expert Systems; Fusion of Fuzzy System and Neural Networks; Fusion of Fuzzy System and Genetic Algorithms
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to interpret the systems which include fuzziness within the scope of fuzzy set theory.
2	Students will be able to combine the information of decision theory and the information of fuzzy set theory.
3	Students will be able to improve the proof techniques of fuzzy set theory.
4	Students will be able to solve problems that include uncertainty with using fuzzy set theory.
5	Students will be able to combine Fuzzy systems and artificial intelligence techniques.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Fuzzy Sets	Chapter 1 / Pages 1-26
2	The Operation of Fuzzy Sets	Chapter 2 / Pages 27-52
3	Fuzzy Relation and Composition	Chapter 3 / Pages 53-90
4	Fuzzy Graph and Relation	Chapter 4 / Pages 91-128
5	Fuzzy Numbers	Chapter 5 / Pages 129-152
6	Fuzzy Functions	Chapter 6 / Pages 153-170
7	Probability and Uncertainty	Chapter 7 / Pages 171-192
8	Midterm 1 / Practice or Review	Chapter 8 / Pages 193-216

9	Review and applications	Chapters 1-6 / Pages 1-170
10	Fuzzy Inference	Chapter 9 / Pages 217-252
11	Fuzzy Control & Fuzzy Expert Systems	Chapter 10 / Pages 253-284
12	Fusion of Fuzzy System and Neural Networks	Chapter 11 / Pages 285-308
13	Review and applications	Chapters 7-10 / Pages 171-276
14	Fusion of Fuzzy System and Genetic Algorithms	Chapter 12 / Pages 309-324
15	Final	Chapters 1-12 / Pages 1-324
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	12	4	48
Special Course Internship (Work Placement)			
Homework Assignments	2	30	60
Quizzes/Studio Critics			
Project			
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	2	30	60
Final (Examination Duration + Examination Prep. Duration)	1	35	35
		Total Workload	245
		Total Workload / 30(h)	8.17
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Big Data Processing and Analytics	BLM4120	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Mehmet Sıddık Aktaş
Instructor(s)	Mehmet Sıddık Aktaş
Asistant(s)	

Course Objectives	This course will offer to students programming models, algorithms and tools of big data computing to support data-intensive applications. Students will get to know the latest research topics of big data platforms.
Course Content	In this course, new computing paradigms that are emerging for big data applications will be covered. These include big data algorithms, big data programming paradigms and platforms, big data analysis tools. In addition, the course will cover a lot of scientific papers.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will learn recent research trends and special topics in big data computing area.
2	Big Data Processing and Analysis concepts will be learned.
3	Big Data Processing and Analysis skills for developing data-intensive applications will be acquired.
4	Big Data computing platforms will be analyzed, and the skills to evaluate (performance, scalability, usability criteria) big data computing applications will be gained.
5	Students may have information about current software used in Big Data Processing.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Course	
2	Map-Reduce Programming Model	
3	Finding Similar Items in Big Data	
4	Mining Data Streams	
5	Link Analysis in Big Data	
6	Frequent Itemsets Mining in Big Data	
7	NoSQL Databases	

8	Midterm 1 / Practice or Review	
9	Recommendation Systems for Big Data	
10	Content and Collaborative Filtering Recommender Systems for Big Data	
11	Dimensionality Reduction	
12	Large-Scale Machine Learning	
13	Mining Social-Network Graphs	
14	Student Presentations	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	6	15
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	6	84
Special Course Internship (Work Placement)			
Homework Assignments	6	5	30
Quizzes/Studio Critics			

Project	1	50	50
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	13	13
Final (Examination Duration + Examination Prep. Duration)	1	13	13
		Total Workload	232
		Total Workload / 30(h)	7.73
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Distributed Systems	BLM4760	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Mehmet Sıddık Aktaş
Instructor(s)	
Asistant(s)	

Course Objectives	Objective in this course is to introduce the main concepts in distributed systems and applications, teach identifying properties of distributed systems and help students gain experiences in designing new distributed applications.
Course Content	In this course, we cover main concepts and objectives of distributed systems. Resource sharing in the web. Types and architectures of distributed systems. Layered arhitecture and middleware. Threads and processes. Client and server architecture. Interprocess communication. Remote Procedure Call. Naming Services. Security in distributed systems. Distributed File system and logical clock
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students can describe main concepts and major architectural styles used in distributed systems.
2	Students can design various processes and use sockets to implement interprocess communication
3	Students can explain the importance, structure and steps of remote procedure call.
4	Students can explain the functions of naming services and design a name server
5	Students can evaluate and design operations in distributed file system.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction, definitions and concepts	Introduction, Ch1
2	DS properties and System Architectures	System Architectures, Ch2
3	Processes and threads	Ch 3
4	Interprocess Communication, sockets, server socket	Interprocess Communication, Ch4 (2nd text)
5	Interactions of Client and server processes	
6	Layered Architecture	

7	Middleware: RPC and RMI	Ch4
8	Midterm 1 / Practice or Review	Ch5
9	Naming	Ch5
10	Data Replication ad Consistency	Ch 7
11	Distributed File Systems	Ch 11
12	Distributed File Systems: Architectures	Ch 11
13	Sample Distributed File Systems	Ch 11
14	Security in Distributed Systems	Ch 9
15	Final	Lecture Notes
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	6	15
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	6	84
Special Course Internship (Work Placement)			
Homework Assignments	6	5	30
Quizzes/Studio Critics			

Project	1	50	50
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	13	13
Final (Examination Duration + Examination Prep. Duration)	1	13	13
		Total Workload	232
		Total Workload / 30(h)	7.73
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Behavior Science	ISL1622	3	5	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Business Administration
Course Coordinator	Turhan Erkmen
Instructor(s)	Turhan Erkmen, Serdar Bozkurt, Aygül Turan
Asistant(s)	Selahaddin Şamil Fidan

Course Objectives	Principles of individual and group behavior, to understand and interpret both type of behavior in the light of such issues as motivation, learning, perception, personality, culture and so on. Related to these objectives, group discussions and case studies will take place in the class.
Course Content	The content and principles of Individual behavior and group behavior with the factors affecting both are examined in relation to the sciences of Psychology and Sociology.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students have ability to be able to comprehend the factors that influence behavior
2	Students discuss to understand how behavior can be altered
3	Student have ability how behavioral problems can be solved.
4	Students investigate and gain skills about human behavior in organizations

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	The Meaning and Content of Behavioral Sciences and its Relationship to other Disciplines	Şimşek, Akgemci, Çelik, 2003, 1-15
2	Learning Principles and Theories	Şimşek, Akgemci, Çelik, 2003, 96-116
3	Perception and Social Perception	Şimşek, Akgemci, Çelik, 2003, 93-96
4	Motivation and Theories of Motivation	Şimşek, Akgemci, Çelik, 2003, 129-154
5	Personality Traits and Theories	Şimşek, Akgemci, Çelik, 2003, 73-90
6	Personality Factors	Şimşek, Akgemci, Çelik, 2003, 80-90

7	Communication and Communication in Organizations	Quick ve Nelson, 2011, 238-272
8	Midterm 1 / Practice or Review	NA
9	The Formation and Components of Attitudes and Attitude Change	Şimşek, Akgemci, Çelik, 2003, 51-71
10	Measurement of Attitudes	Şimşek, Akgemci, Çelik, 2003, 60-71
11	The components, Characteristics and Classification of Culture	Şimşek, Akgemci, Çelik, 2003, 27-48
12	Cultural Variation And Attitudes Toward Culture	Şimşek, Akgemci, Çelik, 2003, 27-48
13	Role, Status and Determinants of Status	Şimşek, Akgemci, Çelik, 2003, 17-26
14	The Characteristics and Types of Groups	Şimşek, Akgemci, Çelik, 2003, 157-160
15	Final	Şimşek, Akgemci, Çelik, 2003, 161-173
16	Final Exam	NA

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	30
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	13	3	39
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	1	20	20

Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	25	25
		Total Workload	146
		Total Workload / 30(h)	4.87
		ECTS Credit	5

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Circuit Theory and Electronical Devices	BLM1033	4	6	3	0	2

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	Hamza Osman İlhan, Erkan Uslu
Asistant(s)	Burak Ahmet ÖZDEN, Emre PARLAK, Elif AŞICI, Ömer Mutlu Türk KAYA

Course Objectives	This course is intended to provide fundamental informations about circuit elements and solution techniques and to apply these informations in the laboratory environment. Also, to teach diode, BJT and FET transistors and OP-AMP structures which are basic electronic elements and to gain ability of analysis and synthesis of circuits formed with these elements.
Course Content	Electric circuit variables; Circuit elements; Resistive circuits, Methods of analysis of resistive circuits, Circuit theorems, The operational amplifier, Energy storage elements, The complete response of first order circuits, The complete response of second order circuits, Complex algebra, Diode; Diode Applications; Diode Logic; Bipolar Junction Transistor (BJT); DC Analysis of BJT Circuits; Diode Transistor Logic (DTL); Resistor Transistor Logic (RTL); Transistor Transistor Logic (TTL); Operational Amplifier (OP-AMP) and Applications; Field Effect Transistor (FET); Junction Field Effect Transistor (JFET); Metal Oxide Semiconductor Field Effect Transistor (MOSFET); DC Analysis of field effect transistor circuits; MOS Logic.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students comprehend basic circuit theory.
2	Students know the behavior of energy storing circuit elements and how to analyze first and second order circuits.
3	Students know how to analyze first and second order circuits.
4	Students have general knowledge about diode, BJT, FET and OP-AMP structures, learn DC and AC analysis methods.
5	Students learn DC and AC analysis methods in electronic circuits.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation

1	Introduction to circuit theory / Charge, current, voltage and power expressions and relationships between them / Types of circuits and circuit elements, basic circuit elements (source, resistor etc.) symbols and the meaning of these symbols / Ohm's law / Mathematical model of the resistor	Hayt pp. 9-38
2	Kirchhoff's current and voltage laws / Analysis of a single-loop and a single-node-pair circuits / Resistance and source combination, series and parallels combinations of the circuit elements / Voltage and current division	Hayt pp. 39-78
3	Nodal analysis / Super node / Mesh analysis / Super mesh	Hayt pp. 79-122
4	Linearity and Superposition Theorem / Source transformations / Thévenin's and Norton's theorems and applying to resistive circuits / Maximum power theorem / Quiz 1	Hayt pp. 123-174
5	Properties of Operational Amplifiers (Op-Amps) / Inverting and Non-inverting Op-Amp circuits / Summing and difference Op-Amp circuits / Cascade connectin of Op-Amps / Model of real Op-Amps	Hayt pp. 175-216
6	Energy storage inductance and capacitance elements, their mathematical models / Series-parallel combinations of energy storage elements / The voltage-current relation of energy storage elements / The DC behavior of energy storage elements.	Hayt pp. 217-260
7	Source-free RL and RC circuits / The application of the unit-step forcing function / Switching concept / RL and RC circuit analysis	Hayt pp. 261-320
8	Midterm 1 / Practice or Review	
9	RLC Circuits / Overdamped Circuit Solution / Critically Damped Circuit Solution / Underdamped Circuit Solution	Hayt pp. 321-370
10	Semiconductor physics, p-n junction, ideal diode, diode characteristics, Equivalent diode models, clipper and clamping circuits, Rectifier circuits, zener diodes, diode logic	Electronic Devices and Circuit Theory Chapter 1, Chapter 2
11	Bipolar junction transistor (BJT), analysis of BJT characteristics, DC analysis of BJT circuits	Electronic Devices and Circuit Theory Chapter 3, Chapter 4
12	Operational amplifier (OP-AMP): comparator, inverting and non-inverting circuits, Operational amplifier: addition, subtraction, derivative and integration circuits	Electronic Devices and Circuit Theory Chapter 14, Chapter 15
13	Field effect transistors (FET), Junction field effect transistor (JFET), DC analysis of FET circuits	Electronic Devices and Circuit Theory Chapter 5, Chapter 6
14	Metal oxide semiconductor field effect transistor (MOSFET) structures, MOS logic	Electronic Devices and Circuit Theory Chapter 5, Chapter 6
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	8	20
Application		
Field Work		

Special Course Internship (Work Placement)		
Quizzes/Studio Critics	2	10
Homework Assignments		
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory	8	2	16
Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics	2	5	10
Project	1	15	15
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	25	25
Final (Examination Duration + Examination Prep. Duration)	1	25	25
		Total Workload	185
		Total Workload / 30(h)	6.17
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Natural Language Processing	BLM4580	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Banu Diri
Instructor(s)	Banu Diri
Asistant(s)	

Course Objectives	Introduce Natural Language and its applications and show its possible applications/realizations and associated constraints.
Course Content	Morphological Analysis; Syntax Analysis; Grammer and Languages; Regular Language; Word Processing Algorithms; Machine Learning; Text Classification; Information Extraction; Information Retrieval; Question Answering Systems; Collacation
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will know the convenience of using Natural Language in our ordinary life.
2	The student will learn the algorithms and methods on the Natural Language Processing domain.
3	The students will recognize and use the domestic and foreign tools.
4	The student will learn all the concept about Natural Language Processing.
5	The student will do a project using the learned material in the class.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Natural Language Processing	Foundations of Statistical Natural Language Processing Section 1
2	Linguistic Essentials	Foundations of Statistical Natural Language Processing Section 3
3	Grammer and Languages	Foundations of Statistical Natural Language Processing Section 11
4	Morphological Anaysis	Foundations of Statistical Natural Language Processing Section 10
5	Syntactic Analysis	Foundations of Statistical Natural Language Processing Section 10

6	Regular Language-RegEx	Handbook of Natural Language Processing, R. Dale, H. Moisl, H.Somers, Marcel Dekker
7	Machine Learning	Introduction to Machine Learning, MIT
8	Midterm 1 / Practice or Review	Introduction to Machine Learning, MIT
9	Deep Learning	Internet Sources
10	Text Categarization	Foundations of Statistical Natural Language Processing Section 16
11	Information Extraction	Foundations of Statistical Natural Language Processing Section 15
12	Information and Retrieval	Foundations of Statistical Natural Language Processing Section 15
13	Question and Answering Systems	Speech and Language Processing Section 23
14	Collocations	Foundations of Statistical Natural Language Processing Section 5
15	Final	Papers
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	10
Presentations/Jury		
Project		
Seminar/Workshop	1	30
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42

Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	3	42
Special Course Internship (Work Placement)			
Homework Assignments	2	20	40
Quizzes/Studio Critics			
Project			0
Presentations / Seminar	1	40	40
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	35	35
		Total Workload	229
		Total Workload / 30(h)	7.63
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
File Organization	BLM3750	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Utku Kalay
Instructor(s)	M. Utku Kalay
Asistant(s)	

Course Objectives	Understanding the Basics of Disk Storage Unit and Disk-based Access Methods.
Course Content	Disc Basics: Performance and Reliability; File Types/Access Types; Static Hash Files; Extendible Hash Files; Linear Hash Files; Multi-level Indexing; ISAM; B-tree; Access Methods on Multiple Attributes: kd-tree; Grid-file
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will have knowledge about disk, disk access performance, and data organization on the disk.
2	The student will have the knowledge about the identification and appropriateness of the data structures needed by different applications evaluation.
3	Student can analyse and compare the disk-based data stuructures like B-tree and hashing.
4	Student will be able to compare the disk-based structures with the in-memory data structures.
5	Student will have the kowledge of the importance of file structures in the dbms implementation.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction and main concepts of storage unit	File Organization and Processing Ch 1,2
2	The fundementals of disk and performance analysis	Fundementals of Database Systems, Ch 13
3	Disk performance and Realiability	Fundementals of Database Systems, Ch 13
4	Field and record organization and record access and reclaiming space	Fundementals of Database Systems, Ch 13
5	Static Hash Files	Fundementals of Database Systems, Ch 10

6	Extendible Hash Files	Fundementals of Database Systems, Ch 10
7	Extendible Hash Files	Fundementals of Database Systems, Ch 10
8	Midterm 1 / Practice or Review	File Organization and Processing Bölüm 4
9	Multi-level indexing	Fundementals of Database Systems, Ch 9
10	Indexing, Index types	Fundementals of Database Systems, Ch 14
11	Multi-level indexing: B-tree and its variations	Fundementals of Database Systems, Ch 14
12	B-tree and its variations	Fundementals of Database Systems, Ch 9
13	Kd-tree	Fundementals of Database Systems, Ch 12
14	Grid-file	Fundementals of Database Systems, Ch 12
15	Final	exam prepFundementals of Database Systems, Ch 12
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	10
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**

Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	12	6	72
Special Course Internship (Work Placement)			
Homework Assignments	1	20	20
Quizzes/Studio Critics			
Project	1	20	20
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	30	60
Final (Examination Duration + Examination Prep. Duration)	1	30	30
		Total Workload	244
		Total Workload / 30(h)	8.13
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Physics 1	FIZ1001	4	6	3	0	2

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Physics
Course Coordinator	Nursel Can
Instructor(s)	
Asistant(s)	

Course Objectives	To introduce the fundamental principles and concepts of physics in detail at freshmen level. To build a strong background for physics major as well as showing the necessity and importance of physics for other branches of natural sciences and engineering through applications in real life, and industry and technology.
Course Content	Physics and Measurement, Vectors, Motion in One Dimension, Motion in Two Dimensions, The Laws of Motion, Circular Motion and Other Applications of Newton’s Laws, The Energy of a System, Conservation of Energy, Linear Momentum and Collisions, Rotation of a Rigid Object About a Fixed Axis, Angular Momentum, Static Equilibrium, Oscillations Motion.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Uses appropriate dimensions and units in solving mechanical problems.
2	Lists the definitions necessary to analyze mechanical systems.
3	It defines the type of movement and calculates the basic quantities related to the movement.
4	Analyzes moving systems in detail using the appropriate coordinate system.
5	It defines the basic quantities that cause movement and calculates them correctly.
6	Establishes appropriate mathematical models to analyze structures with continuous mass distribution and makes solutions appropriate to the distribution geometry.
7	Compares dynamic and/or energy methods to analyze motion.
8	Decomposes multi-particle systems into their components and writes the equations of motion that describe each component.
9	It measures basic quantities experimentally, draws graphs, collects data, tabulates the collected data, and compares experimental and theoretical results.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation

1	Series Expantion, Differential Calculus, Integral Calculus, Physics and Measurements, Dimentional Analysis, Conversion of Units, Estimates and Order-of-Magnitude Calculations, Significant Figures, Coordinate Systems, Vector and Scalar Quantities, Some Properties of Vectors, Components of a Vector	Appendix-B (B.5-7), Textbook Ch1: PHYSİCS AND MEASUREMENT, Textbook Ch3: VECTORS,
2	Unit Vectors, The Scalar Product of Two Vectors, The Vector Product, Position, Velocity and Speed, Instantaneous Velocity and Speed, Partical Under Constant Velocity, Acceleration, Motion Diagrams	Textbook Ch3: VECTORS, Textbook Ch2: MOTION in ONE DIMENSION
3	Partical Under Constant Acceleration,Freely Falling Object, The Position, Velocity and Acceleration Vectors, Two-Dimensional Motion with Constant Acceleration, Projectile Motion, Particle in Uniform Circular Motion, Tangential and Radial Acceleration, Relative Velocity and Relative Acceleration, Experiment 1: The Analysis of an Experiment	Textbook Ch2: MOTION in ONE DIMENSION, Textbook Ch4: MOTION IN TWO DIMENSIONS, Experiment 1
4	The Concept of Force, Newton's First Law and İnertial Frames, Mass, Newton's Second Law, The Gravitational Force and Weight, Newton's Third Law, Forces of Friction, Experiment 1: The Analysis of an Experiment	Textbook Ch5: THE LAWS of MOTION, Experiment 1
5	Extending the Particle in Uniform Circular Motion Model, Nonuniform Circular Motion, Motion in Accelerated Frames, Experiment 2:Newton's Laws of Motion	Textbook Ch6: CIRCULAR MOTION and OTHER APPLICATIONS of NEWTON'S LAWS, Experiment 2
6	Systems and Environments, Work Done by a Constant Force, Work Done by a Varying Force, Kinetic Energy and The Work-Kinetic Energy Theorem, Potential Energy of a System, Conservative and Nonconversative Forces, Relationship Between Conservative Forces and Potential Energy, Energy Diagrams and Equilibrium of a System, Experiment 2:Newton's Laws of Motion	Textbook Ch7: ENERGY of a SYSTEM, Experiment 2
7	Nonisolated System(Energy), Isolated System(Energy), Situations Involving Kinetic Friction, Changes in Mechanical Energy for Nonconservative Forces, Power	Textbook Ch8: CONSERVATION of ENERGY
8	Midterm 1 / Practice or Review	
9	Linear Momentum, Isolated System(Momentum), Nonisolated System(Momentum), Collisions in One Dimension, Collisions in Two Dimensions, Experiment 3: Conservation of Linear Momentum	Textbook Ch9: LINEAR MOMENTUM and COLLISIONS, Experiment 3
10	Collisions in Two Dimensions, The Centre of Mass, Systems of Many Particles, Experiment 3: Conservation of Linear Momentum	Textbook Ch9: LINEAR MOMENTUM and COLLISIONS, Experiment 3
11	Angular Position, Velocity and Acceleration, Rigid Object Under Constant Angular Acceleration, Angular and Translational Quantities, Torque, Rigid Object Under a Net Torque, Calculation of Moments of Inertia, Experiment 4: Moment of Inertia	Textbook Ch10: ROTATION of a RIGID OBJECT ABOUT a FIXED AXIS, Experiment 4
12	The Parallel Axis Theorem, Rotational Kinetic Energy, Energy Considerations in Rotational Motion, Rolling Motion of a Rigid Object, Experiment 4: Moment of Inertia	Textbook Ch10: ROTATION of a RIGID OBJECT ABOUT a FIXED AXIS, Experiment 4
13	The Torque Vector, Nonisolated System(Angular Momentum), Angular Momentum of a Rotating Rigid Object, Isolated System Angular Momentum, Experiment 5: Simple Pendulum and Spring Pendulum	Textbook Ch11: ANGULAR MOMENTUM, Experiment 5

14	Rigid Object in Equilibrium, More on the Center of Gravity, Examples of Rigid Object in Static Equilibrium, Motion of an Object and Attached to a Spring, Particle in Simple Harmonic Motion, Experiment 5: Simple Pendulum and Spring Pendulum	Textbook Ch12: STATIC EQUILIBRIUM, Textbook Ch15: OSCILLATORY MOTION, Experiment 5
15	Final	Textbook Ch15: OSCILLATORY MOTION
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	5	0
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	60
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	14	2	28
Application			0
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments			0
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	15	30
Final (Examination Duration + Examination Prep. Duration)	1	20	20

Total Workload	176
Total Workload / 30(h)	5.87
ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
General Internship	BLM3002	0	3	0	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Banu Diri
Instructor(s)	Banu Diri
Asistant(s)	

Course Objectives	The purpose of the internship is improve the practical works of student in the academical area.
Course Content	Internship jobs in any public or private sector, six weeks (30 working days) requires the acquisition of professional experience. Students who successfully complete the internship are required to follow the rules of the Department of Computer Engineering Internship directive.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will apply the acquired theoretical knowledge in practice.
2	The student, who will offer him a job in the IT field in the future to establish a relationship with the employees learn.
3	The student will assess the student's ability to apply discipline-related knowledge to the field.
4	The student will learn to present the information acquired in an official report.
5	Students will learn to take responsibilities and learn to work with different groups.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Professional Experience	Professional Experience
2	Professional Experience	Professional Experience
3	Professional Experience	Professional Experience
4	Professional Experience	Professional Experience
5	Professional Experience	Professional Experience
6	Professional Experience	Professional Experience
7	Professional Experience	Professional Experience
8	Midterm 1 / Practice or Review	

9	Professional Experience	Professional Experience
10	Professional Experience	Professional Experience
11	Professional Experience	Professional Experience
12	Professional Experience	Professional Experience
13	Professional Experience	Professional Experience
14	Professional Experience	Professional Experience
15	Final	Professional Experience
16	Final	

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	1	10
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury	0	0
Project	1	50
Seminar/Workshop		
Mid-Terms		
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	0	0	0
Laboratory			
Application	0	0	0
Field Work			
Study Hours Out of Class	0	0	0
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	50	50
Presentations / Seminar	0	0	0

Mid-Terms (Examination Duration + Examination Prep. Duration)			
Final (Examination Duration + Examination Prep. Duration)	1	50	50
		Total Workload	100
		Total Workload / 30(h)	3.33
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Operating System Architectures for Large Scale Computing Systems	BLM4500	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Hamza Osman İlhan
Instructor(s)	Hamza Osman İlhan
Asistant(s)	

Course Objectives	This course aims at teaching the students the hardware and software architecture of large scale computing systems
Course Content	Architectural design of large scale computing systems, operating systems for large scale systems
Recommended Optional Program Components	None

Course Learning Outcomes

1	This course enables students to learn the basic concepts and evolution of mainframe systems.
2	This course enables students to learn the principles of operating systems designed for large scale systems.
3	This course enables students to understand the reliability, serviceability and manageability of large scale systems and supporting operating systems.
4	This course enables students to configure and customize operating systems for large scale systems.
5	This course enables students to understand transaction processing capabilities of large scale systems and the operating systems support.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to large scale operating systems and their architecture	
2	Highly available hardware architectures for large scale operating systems	
3	Single-processing, multi-processing and clustering in large scale systems	
4	Software architecture of large scale operating systems	
5	Primary storage management mechanisms for large scale operating systems	

6	Processor and process management mechanisms for large scale operating systems	
7	Load sharing and performance management mechanisms for large scale operating systems	
8	Midterm 1 / Practice or Review	
9	Midterm Exam 1	
10	Application design and implementation on large scale operating systems	
11	Common user interfaces for large scale operating systems	
12	Security mechanism and user tools for large scale operating systems	
13	Security mechanism and user tools for large scale operating systems	
14	Communication mechanism for large scale operating systems	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			

Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	5	10	50
Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	35	35
		Total Workload	239
		Total Workload / 30(h)	7.97
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Embedded Systems	BLM4021	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Ali Can Karaca
Instructor(s)	Ali Can Karaca
Asistant(s)	

Course Objectives	Gaining ability to create solutions based on embedded systems.
Course Content	Introduction to embedded systems and applications Microcontrollers, microprocessors, DSP, FPGA, and ASIC notions 16, 32 and 64 bit microcontrollers (PIC & MSP430 families) ARM and RISC-V architectures ARM instruction set and ARM assembly Raspberry Pi and its versions Data acqusitions; sensors, analog digital converters, digital to analog converters, data processing Timers, interrupts, PWM and DMA on embedded systems Communication hardware and methods Introduction to fundamental concepts of Real Time Systems Real Time Operating Systems Embedded system design applications, Internet of things
Recommended Optional Program Components	None

Course Learning Outcomes

1	The ability of developing application for embedded systems with resource constraints
2	Learning of software development techniques for embedded systems
3	Application development for embedded systems with high level languages
4	Learning of fundamental concepts of Real Time Systems
5	The ability of developing embedded systems with Real Time constraints

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to embedded systems and its applications	
2	Microcontrollers, microprocessors, DSP, FPGA, and ASIC notions	
3	16, 32 and 64 bit microcontrollers (PIC & MSP430 families)	
4	ARM and RISC-V architectures	
5	ARM instruction set and ARM assembly -1	
6	ARM instruction set and ARM assembly -2	

7	Raspberry Pi and its versions	
8	Midterm 1 / Practice or Review	
9	Data acqusitions; sensors, ADC and DAC	
10	Introduction to fundemantal concepts of Real Time Systems	
11	Timers, interrupts, PWM and DMA on embedded systems	
12	Communication hardware and methods	
13	Introduction to fundamental concepts of Real Time Systems	
14	Internet of things	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	30
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	5	70
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			

Project	1	60	60
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	40	40
Total Workload			242
Total Workload / 30(h)			8.07
ECTS Credit			8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Image Processing	BLM4540	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Elif Karslıgil
Instructor(s)	M. Elif Karslıgil
Asistant(s)	

Course Objectives	The aim of this course is to teach students principle techniques and algorithms of image processing
Course Content	Digital Image Fundamentals, Image Enhancement Techniques, Spatial Filtering, Color Image Processing, Image Segmentation, Morphological Image Processing, Texture Analysis, Image Representation and Description, Image Compression, Motion Analysis, Pattern Recognition, Deep Learning for Image Processing Applications
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will understand fundamental underlying principles of digital image processing.
2	Student will learn the basics theory of image processing(hardware and software, digitization, enhancement and restoration, encoding, segmentation, feature detection etc.)
3	Student will understand how to apply such knowledge on real examples of image processing tasks.
4	Student will analyze and implement image processing algorithms.
5	Student will complete a project, write report and present in class on a topic in image processing.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction	
2	Digital Image Fundamentals	
3	Image Enhancement Techniques	
4	Spatial Filtering	
5	Color Image Processing	
6	Image Segmentation 1	
7	Image Segmentation 2	

8	Midterm 1 / Practice or Review	
9	Morphological Image Processing	
10	Image Representation and Description	
11	Motion Anaysis	
12	Pattern Recognition	
13	Deep Learning for Image Processing Applications 1	
14	Deep Learning for Image Processing Applications 2	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	5	20
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	2	40
Final	1	30
	Percentage of In-Term Studies	70
	Percentage of Final Examination	30
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	3	42
Special Course Internship (Work Placement)			
Homework Assignments	5	20	100
Quizzes/Studio Critics			
Project	1	30	30

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	15	30
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	250
		Total Workload / 30(h)	8.33
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Theory of Computation	BLM2502	3	6	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oğuz Altun
Instructor(s)	Oğuz Altun, H.İrem Türkmen
Asistant(s)	

Course Objectives	The objectives of this course are not only to introduce students to the mathematical foundations of computation including automata theory and the theory of formal languages and grammars; but also to provide students with an understanding of such basic concepts as automata, the equivalent regular expressions, the equivalence of languages described by automata, regular expressions, pushdown automata, the equivalent context free grammars, the equivalence of languages described by pushdown automata, context free grammars, Turing machines and the equivalence of languages described by Turing machines.
Course Content	Mathematical Tools (Definitions, Theorems, and Proofs); Types of Proofs; Regular Languages; Finite Automata; Nondeterminism; Regular Expressions; Nonregular Languages; Context-Free Languages; Context-free Grammars; Pushdown Automata; Non-context-free Languages; Turing Machines; Variants of Turing Machines; Definition of "Algorithm"; Decidability; Decidable Languages; Reducibility; NP-completeness; Reducibility; Recognizability
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to analyze finite automata, deterministic and non-deterministic automata, regular expressions, pushdown automata, Turing machines, formal languages, and grammars
2	Students will be able to design finite automata, deterministic and non-deterministic automata, regular expressions, pushdown automata, Turing machines, formal languages, and grammars.
3	Students will be to demonstrate their the understanding of key notions, such as algorithm, computability, decidability, and complexity through problem solving.
4	Students will be familiar with Turing Machines and Problem Classes.
5	Students will improve problem solving skills and gain the ability to prove results of the Theory of Computation

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation

1	Course overview, formation of preliminary concepts, mathematical tools, definitions, theorems, and proofs, types of proofs	
2	Deterministic finite automata (DFA)	
3	Non-deterministic finite automata (NFA)	
4	Equivalence of DFA and NFA, and regular expressions	
5	Epsilon transition, pumping Lemma, pigeonhole principle, and closure properties	
6	Optimal DFA, and overview	
7	Context-free languages, context-free grammars, parse tree, ambiguity, closure properties	
8	Midterm 1 / Practice or Review	
9	Midterm Exam 1	
10	Pushdown automata (PDA)	
11	Overview of context-free grammars, and Church-Turing hypothesis	
12	Turing Machines, Recognition and Computation, Church-Turing Hypothesis	
13	Midterm Exam 2	
14	NP-completeness, decidability, reducibility, and recognizability	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	3	30
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	12	3	36
Laboratory			
Application			
Field Work			
Study Hours Out of Class	6	5	30
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics	2	20	40
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	30	30
Final (Examination Duration + Examination Prep. Duration)	1	44	44
		Total Workload	180
		Total Workload / 30(h)	6.00
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Advanced English 1	MDB1031	3	3	3	0	0

Prerequisite	None

Semester	Fall

Course Language	English
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Modern Languages
Course Coordinator	Burcu Kayarkaya
Instructor(s)	Serap Kartal Yaylı, Burcu Kayarkaya
Asistant(s)	

Course Objectives	• reinforcing students' ability to read analytically • facilitating students' ability to comprehend an appropriately chosen source text to craft an effective analysis • supporting the students' exposure to a specified range of text complexity across a wide range of disciplines aligned to college and career readiness • reinforcing an understanding of relevant words in context and how word choice helps shape meaning and tone
Course Content	• reading strategies • the rhetorical style, the four pillars of argumentation • writing an argumentative essay • vocabulary exercises
Recommended Optional Program Components	None

Course Learning Outcomes

1	Reading a passage and identifying the main points in it
2	Constructing a summary of the important points in the texts
3	Comprehending factual information and distinguishing between facts and ideas in the texts
4	Inferring information from the passage
5	Understanding vocabulary in context

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to the course & Introduction Unit	
2	Unit 1: Rockstar Problems	
3	Unit 1: Rockstar Problems	
4	Unit 2: Six Core Benefits of Personal Development (Language Practice & Writing: Exercises on Paraphrasing)	
5	Unit 3: The Micromobility Revolution (Identifying key points & Chunking)	

6	Unit 3: The Micromobility Revolution (Identifying key points & Chunking)	
7	Putting Strategies In Action & Catch-up	
8	Midterm 1 / Practice or Review	MIDTERM II
9	Unit 4: Is Streaming Killing Cinema Industry? (Summarizing input)	
10	Unit 4: Is Streaming Killing Cinema Industry? (Summarizing cont'd)	--
11	Unit 5: Life Below Water (Summarizing practice)	
12	Unit 5: Life Below Water (Summarizing practice)	
13	Unit 6: Dystopia & In-class Writing Activity	--
14	Unit 6: Dystopia	--
15	Final	
16	FINAL EXAM	NA

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	1	20
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			0
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			

Homework Assignments			
Quizzes/Studio Critics	1	2	2
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	10	10
		Total Workload	92
		Total Workload / 30(h)	3.07
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Advanced English 2	MDB1032	3	3	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Modern Languages
Course Coordinator	Not Assigned
Instructor(s)	
Asistant(s)	

Course Objectives	• reinforcing students’ ability to read analytically • facilitating students’ ability to comprehend an appropriately chosen source text to craft an effective analysis • supporting the students’ exposure to a specified range of text complexity across a wide range of disciplines aligned to college and career readiness • reinforcing an understanding of relevant words in context and how word choice helps shape meaning and tone • encouraging students to write an organized summary of a given text by using annotating and paraphrasing techniques • encouraging students to write a response to a text they have read by making a synthesis of the main points in it.
Course Content	• reading strategies • paraphrasing, summarizing techniques • response writing • vocabulary exercises
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to read a passage by using reading strategies and answer the questions about the text content.
2	Students will be able to understand words in context.
3	Students will be able to construct a summary of the important points in the texts.
4	Students will be able to infer information from the passage.
5	Students will be able to write a response to a text they have read.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Unit 1: The Benefits of Environmental Architecture	
2	Unit 1: The Benefits of Environmental Architecture	
3	Unit 2: Happiness	
4	Unit 2: Happiness	

5	Unit 3: Circadian Rhythm	
6	Unit 3: Circadian Rhythm	
7	Unit 4: Six Ways Social Media is Making you Depressed	
8	Midterm 1 / Practice or Review	Preparing related coursebook
9	Midterm 1	
10	Unit 5: FOMO is Real	
11	Unit 5: FOMO is Real	
12	Unit 6: My Smart Campus	
13	Unit 6: My Smart Campus	
14	Feedback on Midterm Assignment & Catch-up	
15	Final	Preparing related worksheets
16	FINAL EXAM	NA

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	1	20
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			

Homework Assignments			
Quizzes/Studio Critics	1	2	2
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	10	10
		Total Workload	92
		Total Workload / 30(h)	3.07
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Operating Systems	BLM3011	3	5	2	2	0

Prerequisite	BLM1031

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Ahmet Elbir
Instructor(s)	Ahmet Elbir, Ziya Cihan Tayşi
Asistant(s)	

Course Objectives	This course aims at teaching the students the hardware and software architecture of operating systems.
Course Content	Basic architecture of operating systems, hardware and software requirements and application areas of operating systems.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to distinguish different styles of operating system design.
2	Students will understand device and I/O management functions in operating systems as part of a uniform device abstraction.
3	Students will understand the main principles and techniques used to implement processes and threads as well as the different algorithms for process scheduling.
4	Students will understand the main mechanisms used for inter-process communication.
5	Students will understand the main problems related to concurrency and the different synchronization mechanisms available.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Definition of operating system and general concepts of operating system and development history of operating systems	
2	General hardware specifications needed to support operating systems in computer systems	
3	Process concept and basic methods applied for process management	
4	Examination and comparison of process execution and scheduling methods	
5	Ensuring communication and synchronization between processes	

6	Deadlock in processes and solution methods	
7	Memory management, its importance in multi-user systems, methods used to create virtual memory and required hardware specifications	
8	Midterm 1 / Practice or Review	Midterm Exam
9	Methods used for virtual memory creation and required hardware specifications	
10	Introduction of input-output systems and their importance in memory hierarchy	
11	Operating principles of input-output systems, sequential and random access methods	
12	Allocation of input-output systems among users, concept of virtual input-output unit	
13	Introduction of the file system, comparison of flat and hierarchical file systems	
14	Examination of the relationship between the logical mesh system and the physical environment units	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application	1	10
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	10
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28

Laboratory			
Application	14	2	28
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments	1	13	13
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	10	20
Final (Examination Duration + Examination Prep. Duration)	1	5	5
		Total Workload	150
		Total Workload / 30(h)	5.00
		ECTS Credit	5

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Linear Algebra	MAT1320	2	4	2	0	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Mathematics
Course Coordinator	Not Assigned
Instructor(s)	
Asistant(s)	

Course Objectives	
Course Content	
Recommended Optional Program Components	None

Course Learning Outcomes

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1		
2		
3		
4		
5		
6		
7		
8	Midterm 1 / Practice or Review	
9		
10		
11		
12		
13		
14		
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms		
Final		
Percentage of In-Term Studies		0
Percentage of Final Examination		
TOTAL		0

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics	1	15	15
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	15	15
Final (Examination Duration + Examination Prep. Duration)	1	20	20
Total Workload			106
Total Workload / 30(h)			3.53
ECTS Credit			4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Mathematics 1	MAT1071	4	6	3	2	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Mathematics
Course Coordinator	Salih Çelik
Instructor(s)	Pınar Albayrak
Asistant(s)	

Course Objectives	To give information of foundation of Mathematics and to satisfy properties of analytical thinking.
Course Content	Functions:Domain of a function,Functions and Graphs, Even-Odd Functions, Symmetry, Operations on Functions (Sum, difference, multiğlication, division and powers),Composite functions, Piecewise Functions, polynomials and Rational Functions, Trigonometric functions Limits and Contiunity: Limit of a Function and Limit Laws, The Sandwich (The Squeeze theorem), The Precise Definition of a Limit, One-sided Limits, , Limits İnvolving İnfinity, Infinity limits,Contiunityat a point, Continuous Functions, The İntermediate Value Theorem Types of Discontiunity, Differentiation:Tangents ,Normal Lines , The Derivative at a Point, The Derivate as a Function, One-sided Derivatives,Differentiable on an İnterval, , Differentiation Rules, High order Derivatives, Derivatives of Trigonometric Fnctions, The chain rule, Implicit Differentiation, Linearization and Differentials, Increasing Functions and Decrasing Functions,Transcendental Functions:Inverse Functions and Their Derivatives,Logarithms and Exponential Functions and Their Derivatives, Logarithmic Differentiation, Inverse Trigonometric Functions and Their Derivatives, Hyperbolic Functions and Their Derivatives,Inverse Hyperbolic Functions and Their Derivatives,Indeterminate Forms and L'Hospitals Rule, Extrem Values of Functions, Critical Points,Rolle's Theorem, The Mean Value Theorem, The First Derivative Test for Local Extrema, Concavity , The Second Derivative Test for Concavity, Point of İnflection, The Second Derivative Test for Local Extrema,Asymptotes of Graphs Graphing of y=f(x), Antiderivatives, Indefinite İntegrals, Integral table,Integration:Area and Estimating with Finite Sums, Sigma Notation and Limits of Finite Sums, Riemann Sums, Definite İntegral, Properties of Definite İntegral, Area Under the Graph of a nonnegative Function, Average Value of Continuous Functions,Mean Value Theorem fo Definite İntegrals, The Fundamental Theorem of Calculus: Fundamental Theorem Part 1, Fundamental Theorem Part 2, Techniques of Integration: Integration by Substitution, Integration by Parts, Trigonometric Integrals, Reduction Formulas,Trigonometrik Substitutions, Tan (θ/2) subtitutions, Integrations of Rational Functions by Partial Fractions,Applications of definite integrals:Area between two curves, Volumes Using Cross-sections, The Disk Method, the Washer Method, The Ccylindrical Shell method, Arch Length, Areas of Surfuces of Revolution,Improper Integrals, Improper Integrals of Type 1 and Type 2

Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will learn using the concepts of limit, continuity and differetation of one variable functions,
2	Students will learn sketching the graph of a function using asymptotes, critical points and the derivative test for increasing/decreasing and concavity properties,
3	Students will learn setting up and solving max/min problems,
4	Students will learn evaluating definite integrals by using the Fundamental Theorem of Calculus and evaluating areas, volumes and arc lenghts by mean of definit integral,
5	Students will learn applying techniques of integration and working with transcendental functions.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Functions: Domain and Range of a Function, Functions and Graphics, Some Elemanter Functions (Polinomials, Rational Functions, Algebraic Functions), Some Concepts Related to Functions (Even-Odd Functions, Bounded Function, Increasing-Decreasing Functions, Implicit Function), Combinations of Functions ((Sum, Difference, Product and Quotient), Composition of Functions, Piecewise Functions. Transandantal Functions: Trigonometric Functions,	Textbook 1 (Chapter 1)
2	Inverse of a Function, Inverse Trigonometric Functions, Exponential and Logarithmic Functions, Hyperbolic-Inverse Hyperbolic Functions. Limits: Limit of a Function, One-sided Limits, Theorems on Limits (Limit Rules), Squeezing (Pinching) Theorem,	Textbook 1 (Chapter 7,2)
3	Infinite Symbols and Limit. Continuity: Continuity at a Point, Continuity over a Set, Types of Discontinuity, Intermediate Value and Extreme Value Theorems. Derivative: Derivative at a Point, Derivative as a Function, One-sided Derivatives, Derivative over an Interval,	Textbook 1 (Chapter 2,3)
4	Differentiability and Continuity, Differentiation Rules, Higher Order Derivatives, Chain Rule, Implicit Differentiation, Derivatives of Transcendent Functions, Tangent and Normal Lines, Linearization and Differentials.	Textbook 1 (Chapter 3,7)
5	Applications of the Derivative: Rolle and Mean Value Theorems, Increasing-Decreasing Functions, Extremum Values of Functions: First and Second Derivative Tests, Indeterminate Forms, Concavity and Points of Inflection, Asymptotes, Simple Curve Drawings.	Textbook 1 (Chapter 4)
6	Definite Integral: Antiderivatives, Sigma Notation and Limits of Finite Sums, Riemann Sums, Definite Integral, Properties of the Definite Integral, Area of a Region Bounded by a Curve, Mean Value Theorem for Definite Integrals,,	Textbook 1 (Chapter 5)
7	Fundamental Theorem of Calculus (Part 1, Part 2), Indefinite Integral: Integration Tables, Change of Variables in Indefinite-Finite Integrals [Calculus of some trigonometric integrals]. Integration Techniques: Integration by Parts,	Textbook 1 (Chapter 5,8)
8	Midterm 1 / Practice or Review	Textbook 1 (Chapter 4)

9	Trigonometric Substitution, Integrals of Rational Functions (Partial Fractions). Applications of the definite integral: Areas of Plane Regions (Two or more curves).	Textbook 1 (Chapter 6)
10	Volumes of Solids of Revolution: Disk and Washer Techniques, Cylindrical Shell Technique, Arc Length, Areas of Revolution Surfaces.	Textbook 1 (Chapter 6)
11	Imroper Integrals: Integrals of Type I and Type II.	Textbook 1 (Chapter 8)
12	Quiz Parametric and Polar Curves: Polar Coordinates, Relationship between Polar and Cartesian Coordinates, Introduction of Polar Curves (Line, circle and cardioid curves in polar coordinates), Area in Polar Coordinates, Length of a Polar Curve,	Textbook 3 (Chapter 11)
13	Planar Curves and Parametrization, Parametric Derivative, Length of a Parametric Curve. Vectors: Vectors, Dot Product, Angle Between Two Vectors, Perpendicular Vectors, Vector Product, Parallel Vectors,	Textbook 3 (Chapter 2)
14	Lines in Space (Vectorial and parametric equations, Angle between lines), Planes (Vectorial and general equation in space), Angle Between Line and Plane.	Textbook 3 (Chapter 2)
15	Final	
16	Final exam	-

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	1	20
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	5	70

Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	5	70
Special Course Internship (Work Placement)			
Homework Assignments			0
Quizzes/Studio Critics	1	10	10
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	15	15
		Total Workload	185
		Total Workload / 30(h)	6.17
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Mathematics 2	MAT1072	4	6	3	2	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Mathematics
Course Coordinator	Salih Çelik
Instructor(s)	Pınar Albayrak
Asistant(s)	

Course Objectives	To give a broad knowledge and basic understanding of sequences and series and to gain ability of using the concepts of limit, continuity, partial differentiation , double integrals

Course Content	Functions of Several Variables: Functions of Two and Three Variables: Domain and Range, Graphs, Level Curves and Level Surfaces, Limits and Continuity in Functions of Two Variables (Two-Way Test for the Absence of Limit, Continuity of Composite Functions), A Brief Look at Quadratic Surfaces (plane, sphere, ellipsoid, elliptic paraboloid, cylinder, cone), Functions of More Than Two Variables. Partial Derivatives: Partial Derivatives of Functions of Two and Three Variables, Partial Derivative and Continuity, Equality of Second Order Partial Derivatives and Mixed Derivatives, Higher Order Partial Derivatives, Differentiability, Chain Rule for Functions of Two and Three Variables [For Functions with One and Two Independent Variables], Shortly Implicit Derivative [$\partial f/\partial x, \partial f/\partial y$ for $f(x,y)=0$ and $F(x,y,z)=0$, etc.] Directional Derivatives and Gradient Vector: Definition and Calculation of Directional Derivative in the Plane, Gradient Vector, Tangents to Level Curves and Gradients, Directional Derivative in Space, Tangent Planes and Differentials: Tangent Plane and Normal Line of a Surface. Linearizing a Function of Two Variables, Differentials, Extreme Values: Local Extreme Values, Necessary Conditions for Local Extreme Values, Critical and Saddle Points, Second Derivative Test for Local Extreme Values. Multiple Integrals: Double Integrals over Rectangles, Double Integrals as Volumes, Calculating Double Integrals: Fubini's Theorem (First Form), Double Integrals over General Regions, Double Integrals over Bounded Non-Rectangular Regions, Volumes (Volumes between two surfaces), Fubini's Theorem (More Comprehensive Form) Finding Limits of Integration, Using Vertical Sections, Using Horizontal Sections, Properties of Double Integrals, Calculating Area in Double Integrals, Mean Value Theorem. Double Integrals in Polar Form: Finding the limits of integration, Converting Cartesian Integrals to Polar Integrals, Calculating volume using polar coordinates (volume between two surfaces), Transformation of Variables in Double Integrals Midterm Exam Vector-Valued Functions in Space: Definition, Limit and Continuity of Vector-Valued Functions, Derivatives (Velocity and Acceleration Vectors), Arc Length Along a Space Curve, Vector Fields. Curve Integrals: Curve Integral of Vector Fields, Curve Integral with Respect to Coordinates Infinite Sequences: Convergence and Divergence of Sequences (definitions), Calculating Limits of Sequences, Squeeze Theorem for Sequences, Continuous Function Theorem on Sequences, Frequently Encountered Limits, Successively Defined Sequences, Bounded Monotone Sequences, Monotone Sequence Theorem. Infinite Series: Geometric Series, nth Term Test for Divergent Series, Combining Series, Adding or Deleting a Term, Convergence Tests for Series with Positive Terms: Integral Test, p-Series, Harmonic Series, Comparison Test, Limit Comparison Test. Quiz; Ratio and Root Tests. Alternating Series, Alternating Series Test, Absolute and Conditional Convergence. Power Series: Radius of Convergence of a Power Series, Derivative and Integral of Power Series, Taylor and Maclaurin Series: Maclaurin and Taylor Polynomials, Maclaurin and Taylor Series, Applications of Taylor Series: Approximate Value Calculus, Limit Calculus, Calculus of Non-Elementary Integrals.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Student will find to convergence of sequences and series and interval of convergence of power series
2	Student will use vector algebra in 3d-space and in plane and writing the equations of planes and lines
3	Student will understand the limit and the continuity of multivariable functions, computing partial derivatives of them, finding tangent lines, directional derivatives, gradients
4	Student will understand solving the problem of extreme values by using second derivative test

5	Student will evaluate double integrals and applying double integrals for computing of area and volume

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Functions of Several Variables: Functions of Two and Three Variables: Domain and Range, Graphs, Level Curves and Level Surfaces, Limits and Continuity in Functions of Two Variables (Two-Way Test for the Absence of Limit, Continuity of Composite Functions),	Textbook 1 (Chapter 14)
2	A Brief Look at Quadratic Surfaces (plane, sphere, ellipsoid, elliptic paraboloid, cylinder, cone), Functions of More Than Two Variables. Partial Derivatives: Partial Derivatives of Functions of Two and Three Variables, Partial Derivatives and Continuity, Equality of Second Order Partial Derivatives and Mixed Derivatives, Higher Order Partial Derivatives, Differentiability,	Textbook 1 (Chapter 14)
3	Chain Rule for Functions of Two and Three Variables [For Functions with One and Two Independent Variables], Implicit Derivative , Directional Derivatives and Gradient Vector: Definition and Calculation of Directional Derivative in the Plane, Gradient Vector, Tangents to Level Curves and Gradients, Directional Derivative in Space,.	Textbook 1 (Chapter 14)
4	Tangent Planes and Differentials: Tangent Plane and Normal Line to a Surface. Linearization a Function of Two Variables, Differentials, Extreme Values: Local Extreme Values, Necessary Conditions for Local Extreme Values, Critical and Saddle Points, Second Derivative Test for Local Extreme Values.	Textbook 1 (Chapter 14)
5	Multiple Integrals: Double Integrals over Rectangles, Double Integrals by Volume, Evaluation of Double Integrals: Fubini's Theorem (First Form), Double Integrals over General Regions,	Textbook 1 (Chapter 15)
6	Double Integrals over Non-rectangular Bounded Regions, Volumes (volume between two surfaces), Fubini's Theorem (More Comprehensive Figure) Finding the Limits of Integration, Using Orthogonal Sections, Using Horizontal Sections, Properties of Double Integrals, Calculating Area in Double Integrals,.	Textbook 1 (Chapter 15)
7	Mean Value Theorem. Double Integrals in Polar Form: Finding the limits of integration, Converting Cartesian Integrals to Polar Integrals, Area and volume calculations using polar coordinates (volume between two surfaces),	Textbook 1 (Chapter 15)
8	Midterm 1 / Practice or Review	Textbook 1 (Chapter 14)
9	Change of Variables in Double Integrals. Vector Valued Functions in Space: Definition, Limit and Continuity of Vector Valued Functions, Derivatives (Velocity and Acceleration Vectors), Arc Length Along a Space Curve, Vector Fields.	Textbook 1 (Chapter 15,13)
10	Line Integrals: Line Integral of Vector Fields, Line Integral with Respect to Coordinates. Infinite Sequences: Convergence and Divergence of Sequences (definitions), Calculation of Limits of Sequences, Squeeze Theorem for Sequences, Continuous Function Theorem on Sequences,	Textbook 1 (Chapter 16,10)

11	Frequently Encountered Limits, Successively Defined Sequences, Bounded Monotone Sequences, Monotone Sequence Theorem. Infinite Series: Geometric Series, n. Term Test for Divergent Series, Combining Series, Adding or Deleting Term, Convergence Tests for Series with Positive Terms: Integral Test, p Series,	Textbook 1 (Chapter 10)
12	Quiz; Harmonic Series, Comparison Test, Limit Comparison Test. Ratio and Root Tests. Alternating Series, Alternating Series Test,.	Textbook 1 (Chapter 10)
13	Absolute and Conditional Convergence. Power Series: Radius of Convergence of a Power Series, Derivative and Integral of Power Series, Taylor and Maclaurin Series: Maclaurin and Taylor Polynomials, Maclaurin and Taylor Series,	Textbook 1 (Chapter 10)
14	Applications of Taylor Series: Approximation, Limit, Non-Elementary Integrals.	Textbook 1 (Chapter 10)
15	Final	Textbook 1
16	Final	-

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	1	20
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	5	70
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	5	70
Special Course Internship (Work Placement)			

Homework Assignments	0	0	0
Quizzes/Studio Critics	1	10	10
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	15	15
		Total Workload	185
		Total Workload / 30(h)	6.17
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Professional Intership	BLM4002	0	3	0	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Not Assigned
Instructor(s)	Banu Diri
Asistant(s)	

Course Objectives	The purpose of the internship is improve the practical works of student in the academical area.
Course Content	Internship jobs in any public or private sector, six weeks (30 working days) requires the acquisition of professional experience. Students who successfully complete the internship are required to follow the rules of the Department of Computer Engineering Internship directive.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will apply the acquired theoretical knowledge in practice.
2	The student will establish a relationship with the future colleagues who work in the IT field.
3	The student will assess the student's ability to apply discipline-related knowledge to the field.
4	The student will learn to present the information acquired in an official report.
5	The student will learn to take responsibilities and learn to work with different groups.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Professional Experience	
2	Professional Experience	
3	Professional Experience	
4	Professional Experience	
5	Professional Experience	
6	Professional Experience	
7	Professional Experience	
8	Midterm 1 / Practice or Review	

9	Professional Experience	
10	Professional Experience	
11	Professional Experience	
12	Professional Experience	
13	Professional Experience	
14	Professional Experience	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	1	10
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury	0	0
Project	1	50
Seminar/Workshop		
Mid-Terms		
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	0	0	0
Laboratory			
Application	0	0	0
Field Work			
Study Hours Out of Class			
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	50	50
Presentations / Seminar	0	0	0

Mid-Terms (Examination Duration + Examination Prep. Duration)			
Final (Examination Duration + Examination Prep. Duration)	1	50	50
		Total Workload	100
		Total Workload / 30(h)	3.33
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Semiconductor Physics for Engineering	FIZ1951	3	6	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Physics
Course Coordinator	Yusuf Yerli
Instructor(s)	
Asistant(s)	

Course Objectives	
Course Content	
Recommended Optional Program Components	None

Course Learning Outcomes

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1		
2		
3		
4		
5		
6		
7		
8	Midterm 1 / Practice or Review	
9		
10		
11		
12		
13		
14		
15	Final	
16		

Evaluation System		
Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms		
Final		
Percentage of In-Term Studies		0
Percentage of Final Examination		
TOTAL		0

ECTS Workload Table			
Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours			
Laboratory			
Application			
Field Work			
Study Hours Out of Class			
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)			
Final (Examination Duration + Examination Prep. Duration)			
Total Workload			0
Total Workload / 30(h)			0.00
ECTS Credit			0

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Microprocessor Systems and Assembly Language	BLM3061	4	5	3	0	2

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Erkan Uslu
Instructor(s)	Erkan Uslu, Furkan Çakmak
Asistant(s)	

Course Objectives	Ability to use 80x86 assembly language as a low level programming language, to interact with Input-Output devices and interact with high level programming languages. Theory and applications on Intel microprocessors, peripheral devices and memory organizations.
Course Content	Intel 8086 Processor Family Architecture; Registers and Their Functions; Flags;, 80x86 Assembly Mnemonics; Addressing Models; Pseudo Commands; EXE and COM Style Programs; Procedures and Procedure Calls; Macro; Segment Union; Parameter Passing Techniques; Interrupts; Interaction with High Level Programming Languages. Intel 8086 and 286 Architecture; Input-Output Device; 8255 PPI; 8251 USART; 8254 PIT; ADC and DAC; Interrupt Requests; 8259 PIC; Memory Organizations; Address Decoding.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Knows the 80x86 processor's command set in real mode.
2	Being familiar with procedure calls, interrupt and macro handling as well as the methods of passing parameters.
3	Understending the concept of microprocessor architecture and programming.
4	Ability to design hardware microprocessor systems to meet desired needs.
5	Ability to use techniques and modern engineering tools necessary for engineering practice..

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	General specifications of low level programming language, Numbers, number systems and information coding systems. General introduction to 80x86 processor family, structure, architecture, registers sets, flags and segmented organization	
2	Mnemonics (data transfer, arithmetic, branch, loop, flags , logic)	

3	Mnemonics (shift , rotate, string operations, prefixes, pseudo commands, addressing modes)	
4	Tools for low level programming, EXE style programming	
5	COM style programming, procedures and macros	
6	Parameter passing methods, common segment usage	
7	Assembly and high level programming languages integration	
8	Midterm 1 / Practice or Review	
9	Input-Output Device Programming	
10	8255 PPI - Programmable Parallel Interface	
11	8251 USART	
12	8254 Peripheral Interval Timer (PIT)	
13	ADC and DAC Applications	
14	8259 and Interrupt Requests	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	10	24
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	2	14
Homework Assignments	2	8
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	14
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	10	2	20
Application			

Field Work			
Study Hours Out of Class	13	2	26
Special Course Internship (Work Placement)			
Homework Assignments	2	10	20
Quizzes/Studio Critics	2	10	20
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	10	10
		Total Workload	148
		Total Workload / 30(h)	4.93
		ECTS Credit	5

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Mobile Programming	BLM3520	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Distance Learning

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Amaç Güvensan
Instructor(s)	M. Amaç Güvensan
Asistant(s)	

Course Objectives	To teach mobile programming techniques and basics of mobile technologies
Course Content	An Overview of Mobile Technologies ; Mobile Devices ; Mobile OS ; Introduction to Mobile Application Development ; Mobile App Components ; Application Lifecycle ; User Interface Design (Menus, Dialog boxes, etc.) ; RecyclerView ; ViewPager ; ArrayAdapters; Databases on Smartphones and Data Management ; Sensors on Smartphones and Sensor Data Collection ; Broadcast Receivers ; Notifications, User Rights and Permissions ; Location-based Services ; Background Tasks
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will design, develop and test a mobile application taking the smartphone restrictions into account.
2	Students will learn the (dis)advantages of each mobile development methods.
3	Students will follow up-to-date progress on mobile technologies.
4	Students will learn in-situ (on-premise) processing techniques.
5	Students will able to upload his/her mobile application into app market.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	An Overview of Mobile Technologies	
2	Mobile Operating Systems and Mobile Application Programming Languages	
3	Application Development Tools/Environments and Android Architecture	
4	Application Lifecycle, Activities, Intents, and Layouts	
5	User Interface Components, Widgets and Interacting with other Apps	

6	Data Management on Smartphones	
7	Sensors on Smartphones and Sensor Data Collection	
8	Midterm 1 / Practice or Review	
9	Review and applications	
10	Notifications	
11	Background Tasks	
12	Location-based Services	
13	Maps	
14	App Stores and Their Regulations	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	1	5
Homework Assignments	3	15
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	12	3	36
Special Course Internship (Work Placement)			
Homework Assignments	3	24	72

Quizzes/Studio Critics	1	8	8
Project	1	48	48
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	16	16
Final (Examination Duration + Examination Prep. Duration)	1	24	24
		Total Workload	246
		Total Workload / 30(h)	8.20
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction To Game Development	BLM3130	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oğuz Altun
Instructor(s)	Oğuz Altun
Asistant(s)	

Course Objectives	Student should get a review of issues related to developing games and develop at least one simple game.
Course Content	Techniques used in Game development, game engines, game design, play testing.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students have an understanding of how game engines work.
2	Students.have an understanding of basic game design consideratons.
3	Students can develop prototypes of their game ideas.
4	Students have an understanding of game artificial intelligence algorithms.
5	Students have an understanding of development of 3D game models.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction	
2	Game Design Essentials	
3	Game Design Document and Examples	
4	Game Engine Essentials	Gibson Bölüm 15, 16, 18, 24
5	Version Control, Game Example I	
6	Character and Level Design	
7	Game Testing	Schell Bölüm 27
8	Midterm 1 / Practice or Review	
9	Game Testing	Schell Bölüm 27
10	Game AI Essentials	
11	Pathfinding	Madhav Bölüm 9

12	Search Trees	
13	Project Presentations	
14	Project Presentations	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	10
Presentations/Jury		
Project	1	35
Seminar/Workshop		
Mid-Terms	1	15
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	3	39
Special Course Internship (Work Placement)			
Homework Assignments	2	22	44
Quizzes/Studio Critics			
Project	1	40	40
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	52	52
		Total Workload	237

Total Workload / 30(h)	7.90
ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Web Services and Service Oriented Architecture	BLM4110	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Mehmet Sıddık Aktaş
Instructor(s)	
Asistant(s)	

Course Objectives	To learn fundamentals of service oriented programming which is the base of cloud based systems.
Course Content	Web Services, Service Oriented ArchitechtureService Komposition, Business Process Management
Recommended Optional Program Components	None

Course Learning Outcomes

1	Learn Cloud Computing terms.
2	Learn the importance of cloud computing in SOA.
3	Learn Web Service Standards.
4	Be able to select proper web service design metodology
5	Learn Business Process Management.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Basic Definition	
2	Distributed Computing	
3	Service Oriented Architecture	
4	Service Oriented Computing	
5	Fundamental Protocols (SOAP, WSDL, UDDI)	
6	Web Services	
7	Web Service Creation	
8	Midterm 1 / Practice or Review	
9	BPEL – Business Process Execution Language	
10	BPEL – Business Process Execution Language	

11	WSDL - Web Service Description Language	
12	Web Service Life Cycle	
13	Web Service Life Cycle	
14	RDF – Resource Description Framework	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	6	15
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	6	84
Special Course Internship (Work Placement)			
Homework Assignments	6	5	30
Quizzes/Studio Critics			
Project	1	50	50
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	13	13
Final (Examination Duration + Examination Prep. Duration)	1	13	13

Total Workload	232
Total Workload / 30(h)	7.73
ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Robot Technologies	BLM4830	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Furkan Çakmak
Instructor(s)	Sırma Yavuz, Erkan Uslu, M. Fatih Amasyalı, Furkan Çakmak
Asistant(s)	

Course Objectives	Learning basic problems and solution techniques in mobile robotics area.
Course Content	Learning ROS operating system. Mobile robot kinematics, basic methods and applications used in mobile robot problems. Determining whether a method fits into the problem given.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to recognize the basic problems in the aera.
2	Students will be able to construct the suitable models for the problems.
3	Students will be able to determine the proper solution for the models
4	Students will understand the limitations of available tools
5	Students will know how to interpret results

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction of the Course, ROS Platform, Robot Types and Robotics Topics	
2	Introduction of ROS Platform and Examples - 1	
3	Introduction of ROS Platform and Examples - 2	
4	Sensors - Types and Mechanisms	
5	Sensor Applications	
6	Odometry and Localization Concepts	
7	Mapping Methods - 1. Assignment Presentation	
8	Midterm 1 / Practice or Review	Introduction to Autonomous Mobile Robots, Chapter 6.3

9	Mapping Applications	
10	Navigation Methods and Applications - 2nd Assignment Demonstration	
11	Exploration Methods and Applications	
12	Image Processing Techniques on Robots - 3rd Assignment Presentation	
13	3D Mapping Methods	
14	3D Mapping Applications	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	3	30
Presentations/Jury		
Project	1	15
Seminar/Workshop		
Mid-Terms	1	15
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	3	15	45
Quizzes/Studio Critics			0
Project	1	25	25

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	35	35
Final (Examination Duration + Examination Prep. Duration)	1	65	65
		Total Workload	240
		Total Workload / 30(h)	8.00
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Numerical Analysis	BLM1022	3	6	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Banu Diri
Instructor(s)	Banu Diri, Ahmet Elbir
Asistant(s)	

Course Objectives	Explanation of how to solve a complex non-practical problem manually with approximate numerical solutions.
Course Content	Erros and Mistakes, Solution of Nonlinear Equations, Solution of Nonlinear & Linear Systems, Numerical Integration, Numerical Differentiation, Difference Table, Interpolation, Curve Fitting, Solution of Differential Equations which are the main topics of numerical analysis will be explained.
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will learn the methods to solve any advanced mathematical problem which is conventionaly solvable or not.
2	The student will be able to interpret the numerical calculations of the methods and errors.
3	The student will be able to solution of single-variable and non-linear equations.
4	The student will know how to solve the integral and derivative numerical methods.
5	The student will know now to use of numerical interpolation and approximation of functions.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Matematik Modeling and Solving of Mathematic Problems	Mühendisler için Sayısal Yöntemler Section 1-3
2	Number representation, round-off error, truncation error, mistakes	Numerical Methods for Mathematics, Science and Engineering Section 1
3	The Solution of Nonlinear Equations - Close Methods	Numerical Methods for Mathematics, Science and Engineering Section 2
4	The Solution of Nonlinear Equations - Open Methods	Numerical Methods for Mathematics, Science and Engineering Section 2

5	The Solution of Root of Polynomial	Mühendisler için Sayısal Yöntemler Section 2-7
6	The Solution of Linear Systems	Numerical Methods for Mathematics, Science and Engineering Section 3
7	The Solution of Nonlinear Systems	Mühendisler için Sayısal Yöntemler Section 3-12
8	Midterm 1 / Practice or Review	Midterm
9	Numerical Derivative	Numerical Methods for Mathematics, Science and Engineering Section 7
10	Numerical Integration	Numerical Methods for Mathematics, Science and Engineering Section 6
11	Interpolation-1	Numerical Methods for Mathematics, Science and Engineering Section 4
12	Interpolation-2	Numerical Methods for Mathematics, Science and Engineering Section 4
13	Curve Fitting	Midterm ExamNumerical Methods for Mathematics, Science and Engineering Section 5
14	The Solution of Differential Equations	Numerical Methods for Mathematics, Science and Engineering Section 9
15	Final	The projects will be reviewed
16		Final

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	3	42
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	30	60
Final (Examination Duration + Examination Prep. Duration)	1	15	15
		Total Workload	189
		Total Workload / 30(h)	6.30
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Digital Signal Processing	BLM3620	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Ali Can Karaca
Instructor(s)	Ali Can Karaca
Asistant(s)	

Course Objectives	This course is intended to teach how to process digitalized information in digital signal processing systems.
Course Content	Discrete-Time Signals and Systems; Discrete-Time Fourier Transform (DTFT); Discrete Fourier Transform (DFT); Discrete-Time Processing of Continuous-Time Signals; z-Transform; Frequency-Domain Analysis of Linear and Time-Invariant Systems; Digital Filter Design Techniques.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will gain a viewpoint in fundamental principles and techniques of digital signal processing.
2	Students will gain ability for understanding and designing new digital signal processing systems.
3	Students will gain ability to design digital filters with a computer based approach.
4	Students will understand how to apply mathematical concepts to real world problems.
5	Students will be able to develop algorithms for implementing digital signal processing concepts.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction / Signals, Systems, and Signal Processing / Classification of Signals / The Concept of Frequency in Continuous-Time and Discrete-Time Signals	Proakis ch.1 Mitra ch. 1
2	The Sampling Process / Analog-to-Digital and Digital-to-Analog Conversion / Sampling and Reconstruction of Continuous-Time Bandpass Signals	Proakis ch.1,6 Mitra ch. 2
3	Discrete-Time Signals/ Discrete-Time Systems / Analysis of Discrete-Time Linear Time-Invariant systems / Discrete-Time Systems Described by Difference Equations	Proakis ch.2 Mitra ch. 2

4	Implementation of Discrete-Time Systems / Convolution / Correlation of Discrete-Time Signals	Proakis ch.2 Mitra ch. 2
5	Frequency Analysis of Discrete-Time Signals / Frequency-Domain and Time-Domain Signal Properties / Properties of the Fourier Transform for Discrete-Time Signals	Proakis ch.4 Mitra ch. 4
6	Frequency-Domain Characteristics of Linear Time-Invariant Systems / Frequency Response of LTI Systems / Linear Time-Invariant Systems as Frequency-Selective Filters	Proakis ch.5 Mitra ch. 4
7	Frequency Domain Sampling:The Discrete Fourier Transform / Properties of the DFT / Frequency Analysis of Signals Using the DFT	Proakis ch.7 Mitra ch. 5
8	Midterm 1 / Practice or Review	Proakis ch.8 Mitra ch. 5
9	Frequency response of time-invariant systems. Response to real sinusoidal signals. Ideal filters and applications. Time domain and frequency domain.	Chapter 10, James H McClellan, Ronald W. Schaffer and Mark A. Yoder, Signal Processing First & Lecture notes
10	The z-Transform And Its Application To The Analysis Of LTI Systems / Properties of the z-Transform / Inversion of the z-Transform / Analysis of Linear Time Invariant Systems in the z-Domain	Proakis ch.3 Mitra ch. 6
11	Structures for the Realization of Discrete-Time Systems / Structures for FIR Systems / Structures for IIR Systems	Proakis ch.9 Mitra ch. 7,8
12	Design Of Digital Filters / Design of FIR Filters / Design of IIR Filters From Analog Filters	Proakis ch.10 Mitra ch. 9,10
13	Midterm 2	
14	Multirate Digital Signal Processing	Proakis ch.11 Mitra ch. 13
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	5	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60

Percentage of Final Examination	40
TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	13	3	39
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	3	39
Special Course Internship (Work Placement)			
Homework Assignments	2	25	50
Quizzes/Studio Critics			
Project	1	50	50
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	28	28
Final (Examination Duration + Examination Prep. Duration)	1	35	35
		Total Workload	241
		Total Workload / 30(h)	8.03
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Seminar and Professional Ethics	BLM3042	3	3	3	0	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Yunus Emre Selçuk
Instructor(s)	Yunus Emre Selçuk
Asistant(s)	

Course Objectives	Öğrencilerin Bilgisayar Mühendisliği alanında ve eğitimlerine katkı sağlayacakları bir konuda literatür çalışması yapması, araştırma yapması, veri toparlayarak bunları yorumlayabilmesi, sonuçlarını raporlayabilmesi, topluluk önünde sunabilmesine yönelik konular anlatılır. Meslek odalarının ve kanunların belirlediği mesleki ilke ve düzenlemeler tartışma ortamında öğrencilere aktarılır. Derste aktarılan tanımlar ile ilgili öğrenciler araştırma ödev ve projeleri hazırlar ve sunar. Ayrıca yaşam boyu öğrenme, girişimcilik ve sürdürülebilir kalkınma konularında öğrenciler bilgilendirilir.
Course Content	
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will make academic research on a topic.
2	Students will analyze and interpret a research data.
3	Students will prepare their research as report and presentation, speak and share information in front of the community.
4	Students will gain applicable knowledge on occupational regulations and rules of computer engineering.
5	Students gain knowledge to identify the global and societal impacts of their professions and initiatives, and to improve and sustain them.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to lecture and course Explanations	
2	Research methods and techniques, Report preparation and writing rules	
3	Presentation techniques and Determination of seminar topics	
4	UN Sustainable development	

5	Life-long learning	
6	Entrepreneurship, Philosophy of Engineering and Ethics	
7	Chamber of Computer Engineers, Code of Ethics ACM, IEEE Code of Ethics, Chamber of Engineers and Architects Code of Ethics	
8	Midterm 1 / Practice or Review	
9	Intellectual and Industrial Property Rights Law and Other Legalizations, Material and Moral Rights on a Intellectual Property	
10	Program defined by law	
11	Legislation of Publishing on the Internet Law	
12	Trademark and Patents	
13	Presentation	
14	Presentation	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation	14	10
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	10
Presentations/Jury	1	20
Project		
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	1	14

Special Course Internship (Work Placement)			
Homework Assignments	2	4	8
Quizzes/Studio Critics			
Project			
Presentations / Seminar	1	10	10
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	104
		Total Workload / 30(h)	3.47
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
System Analysis and Design	BLM2042	2	3	2	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oya Kalipsiz
Instructor(s)	Oya Kalipsiz, Göksel Biricik
Asistant(s)	

Course Objectives	To teach the concept of systems analysis, information systems and information system development life cycle.
Course Content	System Concept and General System Theory / Information System and Information System Types / Information System Development Process / System Analyst Duties and Capabilities / Preliminary Investigation and Feasibility Analysis / System Proposal Preparation and Presentation / Systems Analysis / Systems Design /Object Oriented Analysis and Design / Systems Implementation / New System to Handle Process
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students know the system concept and implement a system model.
2	Students understand the information system development process.
3	Students apply various structured analysis and design techniques.
4	Students design system solutions that fit the needs of the business.
5	Students understand and practise the responsibilities of the computer professionals on different roles.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to System Analysis and Design, System Concepts	
2	Information System and Information System Types,Information System Development Process	
3	System Analyst Duties and Capabilities, Preliminary Investigation and Feasibility Analysis	
4	Application1: Gantt and PERT diagram	
5	Systems Analysis: Data Collection and Data Modeling	

6	System Design	
7	Object Oriented Analysis	
8	Midterm 1 / Practice or Review	
9	Object Oriented Design	
10	Data Base Design, Systems Implementation, CASE tool	
11	Application2: System Design Modeling Application (Structure Diagram)	
12	Towards to New System	
13	Sistem Bakım ve Desteği	
14	Project Presentations	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	3	15
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	8	2	16
Special Course Internship (Work Placement)			

Homework Assignments			
Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	12	12
Final (Examination Duration + Examination Prep. Duration)	1	12	12
		Total Workload	98
		Total Workload / 30(h)	3.27
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
System Programming	BLM3580	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Göksel Biricik
Instructor(s)	Göksel Biricik
Asistant(s)	

Course Objectives	Obtaining a general knowledge about the technologies used to develop web/Internet applications and to develop a team project.
Course Content	System Programming Concepts; 2-Tier, 3-Tier and N-Tier Application Development Models; Client/Server Architectural Model; HTML; CSS; Scripting; XML; XSLT; DTD; W3C-Schema; DOM; SAX; General Structure of RPC-based Applications; General Structure of RMI-based Applications; Web based Application Development Tools, System Security;
Recommended Optional Program Components	None

Course Learning Outcomes

1	Knowledge on the differences between client-server and conventional applications.
2	Learns HTML , CSS and JavaScript technologies in order to develop web based applications.
3	Learns how to use XHTML and XML for electronic data transfer, XPath and XSLT for document transformations, DTD and XSD used for validity purpose and DOM and SAX to edit XML files.
4	Knows the general structure of RPC, RMI, and Web Services for distributed application development.
5	Gain the ability to develop a Web-based application as a group work.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	System programming concept	Client/Server Survival Guide, Orfali,R., Harkey, D., Edwards, J
2	Client/Server based applications and their specifications	Client/Server Survival Guide, Orfali,R., Harkey, D., Edwards, J
3	2-Tier, 3-Tier and Multi-Tier application specifications.	Internet&World Wide Web: How To Program,Deitel,H.M.,Deitel,P.J., Neito, T.R

4	Web applications, HTML and CSS	Internet&World Wide Web: How To Program,Deitel,H.M.,Deitel,P.J., Neito, T.R
5	Javascript and client side controls	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
6	XML, and DTD to validate XML documents	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
7	Using XSD to validate XML documents	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
8	Midterm 1 / Practice or Review	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
9	XPATH , XSLT Usage	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
10	Introduction to DOM and SAX	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
11	RPC and application development with RPC	Power Programming With RPC, Bloomer, J.
12	RMI and application development with RMI	Java.rmi: Remote Method Invocation Guide, Pitt, E, McNiff K
13	Comparison of distributed application development technologies	
14	Student project presentations	Presentation of developed web application
15	Final	Presentation of developed web application
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	30
Seminar/Workshop		
Mid-Terms	2	30

Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	90	90
Presentations / Seminar	1	10	10
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	25	50
Final (Examination Duration + Examination Prep. Duration)	1	30	30
Total Workload			250
Total Workload / 30(h)			8.33
ECTS Credit			8

Extra Notes	Teams consisting of at least 3 students will be asked to make a web-based term project. Teams formed by students during the third week will choose a project topic that they are interested in among the project topics to be determined by the lecturer. This process will be done by the lecturer for students who do not make a group and / or do not have a topic. The project is expected to be finalized. Project teams will make interviews with the lecturer at 6, 9, and 12 weeks by appointment at extra-curricular times to demonstrate their progress. The produced projects will be evaluated by making presentations in the last two weeks of the semester. Students who do not contribute to the project and / or attend the meeting will be evaluated in different ways.

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
System Programming	BLM3580	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Göksel Biricik
Instructor(s)	Göksel Biricik
Asistant(s)	

Course Objectives	Obtaining a general knowledge about the technologies used to develop web/Internet applications and to develop a team project.
Course Content	System Programming Concepts; 2-Tier, 3-Tier and N-Tier Application Development Models; Client/Server Architectural Model; HTML; CSS; Scripting; XML; XSLT; DTD; W3C-Schema; DOM; SAX; General Structure of RPC-based Applications; General Structure of RMI-based Applications; Web based Application Development Tools, System Security;
Recommended Optional Program Components	None

Course Learning Outcomes

1	Knowledge on the differences between client-server and conventional applications.
2	Learns HTML , CSS and JavaScript technologies in order to develop web based applications.
3	Learns how to use XHTML and XML for electronic data transfer, XPath and XSLT for document transformations, DTD and XSD used for validity purpose and DOM and SAX to edit XML files.
4	Knows the general structure of RPC, RMI, and Web Services for distributed application development.
5	Gain the ability to develop a Web-based application as a group work.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	System programming concept	Client/Server Survival Guide, Orfali,R., Harkey, D., Edwards, J
2	Client/Server based applications and their specifications	Client/Server Survival Guide, Orfali,R., Harkey, D., Edwards, J
3	2-Tier, 3-Tier and Multi-Tier application specifications.	Internet&World Wide Web: How To Program,Deitel,H.M.,Deitel,P.J., Neito, T.R

4	Web applications, HTML and CSS	Internet&World Wide Web: How To Program,Deitel,H.M.,Deitel,P.J., Neito, T.R
5	Javascript and client side controls	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
6	XML, and DTD to validate XML documents	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
7	Using XSD to validate XML documents	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
8	Midterm 1 / Practice or Review	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
9	XPATH , XSLT Usage	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
10	Introduction to DOM and SAX	XML:How to Program, Deitel,H.M., Deitel, P.J., Neito, T.R., Lin, T.M., Sadhu, P.
11	RPC and application development with RPC	Power Programming With RPC, Bloomer, J.
12	RMI and application development with RMI	Java.rmi: Remote Method Invocation Guide, Pitt, E, McNiff K
13	Comparison of distributed application development technologies	
14	Student project presentations	Presentation of developed web application
15	Final	Presentation of developed web application
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	30
Seminar/Workshop		
Mid-Terms	2	30

Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	90	90
Presentations / Seminar	1	10	10
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	25	50
Final (Examination Duration + Examination Prep. Duration)	1	30	30
Total Workload			250
Total Workload / 30(h)			8.33
ECTS Credit			8

Extra Notes	Teams consisting of at least 3 students will be asked to make a web-based term project. Teams formed by students during the third week will choose a project topic that they are interested in among the project topics to be determined by the lecturer. This process will be done by the lecturer for students who do not make a group and / or do not have a topic. The project is expected to be finalized. Project teams will make interviews with the lecturer at 6, 9, and 12 weeks by appointment at extra-curricular times to demonstrate their progress. The produced projects will be evaluated by making presentations in the last two weeks of the semester. Students who do not contribute to the project and / or attend the meeting will be evaluated in different ways.

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Statistical Data Analysis	BLM3590	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Nizamettin Aydın
Instructor(s)	Nizamettin Aydın
Asistant(s)	

Course Objectives	Solving various problems by using basic statistical methods
Course Content	The basic laws of probability and descriptive statistics, conditional probability, random variables, expectation, discrete and continuous probability models, joint and sampling distributions, hypothesis testing, point estimation, confidence intervals, contingency tables, logistic regression, linear and multiple regression.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students recognize the characteristics of the data that can be obtained from surveys.
2	Students create an appropriate data collection tool in a case investigated.
3	Students can prepare available data by using appropriate methods in a case investigated.
4	Students can determine the purpose of the research and data analysis technique in one study.
5	Students can evaluate the research problem based on the existing evidence.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction – Statistical Methods in the Context of Scientific Studies. Sampling. Observational Studies and Experiments. Data Exploration and Analysis. Statistical Inference. Computation Using R.	
2	Data Exploration – Data Visualization and Summary Statistics. Variable Types. Exploring Categorical Variables. Exploring Numerical Variables. Data Preprocessing.	
3	Exploring Relationships – Visualizing and Summarizing Relationships between Variables. Relationships between Two Numerical Random Variables. Relationships between Categorical Variables. Relationships between Numerical and Categorical Variables.	

4	Probability – Probability as a Measure of Uncertainty. Complement, Union, and Intersection. Disjoint Events. Conditional Probabilities. Independent Events. Bayes' Theorem.	
5	Random Variables and Probability Distributions – Random Variables. Probability Distributions. Cumulative Distribution Function and Quantiles.	
6	Estimation – Parameter Estimation. Point Estimation. Sampling Distribution. Confidence Intervals. Margin of Error.	
7	Hypothesis Testing – Hypothesis Testing for the Population Mean. Statistical Significance. Hypothesis Testing using t-tests. Hypothesis Testing for Population Proportion	
8	Midterm 1 / Practice or Review	
9	Analysis of Variance (ANOVA) – Introduction.	
10	Analysis of Variance (ANOVA) – Introduction. The Assumptions of ANOVA.	
11	Analysis of Categorical Variables – Pearson's $\chi 2$ Test for One Categorical Variable. Pearson's $\chi 2$ Test of Independence. Contingency Tables.	
12	Regression Analysis – Linear Regression Models with One Binary Explanatory Variable. Statistical Inference Using Simple Linear Regression Models. Linear Regression Models with One Numerical Explanatory Variable. Model Assumptions and Diagnostics. Multiple Linear Regression.	
13	Clustering – K-means Clustering. Hierarchical Clustering. Standardizing Variables Before Clustering.	
14	Bayesian Analysis – Introduction. Prior and Posterior Probabilities. Bayesian Inference. Estimation. Hypothesis Testing.	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		5
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	4	5
Homework Assignments	3	15
Presentations/Jury	1	5
Project	1	10
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40

Percentage of In-Term Studies	60
Percentage of Final Examination	40
TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	3	15	45
Quizzes/Studio Critics	4	6	24
Project	1	20	20
Presentations / Seminar	1	20	20
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	243
		Total Workload / 30(h)	8.10
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Turkish language 1	TDB1031	0	2	2	0	0

Prerequisite	None

Semester	Fall

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Distance Learning

Owner Academic Unit	Department of Turkish Language
Course Coordinator	Yakup Çelik
Instructor(s)	Hilal TUFAN, Zeliha ÇELEN BOZTİLKİ, Arzuhan KOCABAŞ, Nurullah ARVAS
Asistant(s)	

Course Objectives	Structure of Turkish and acquisition of basic grammar rules, comprehension of reading texts, expanding learners’ vocabulary knowledge.
Course Content	History and basic rules of Turkish language, reading exemplary literary and scientific texts.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Know about the languages used in the world and the place of Turkish among
2	Acquires the correct use of spelling rules and punctuation marks
3	Acquires a larger vocabulary
4	Can use science and knowledge in a better way.
5	Acquires reading habit and pleasure

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction of the content of the the course and references of course.	Textbook
2	Communication.	Textbook
3	Definition of language, characteristics of languages, the relationship between language and culture, relationship between language and nationality. Formal and informal language.	Textbook
4	Languages of the world . Historical development of Turkish and the place of Turkish among the languages of the world.	Textbook
5	Current problems of the Turkish in light of the modern texts. The problems with the spelling of the words in Turkish accompanied by compiled texts.	Article

6	Spelling rules. Punctuation. The importance of the punctuation. Application of punctuation.	Textbook
7	Spelling rules accompanied by contemporary texts.	Column
8	Midterm 1 / Practice or Review	
9	Text analysis: Article	Article
10	Writing	Textbook
11	Writing exercises,text analysis.	Textbook
12	Formal writing styles.	Textbook
13	Expression disorders. Exercises.	Textbook
14	Analysis of expression disorders accompanied by contemporary texts.	Textbook
15	Final	
16	Final exam.	Textbook

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	30
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	13	2	26
Laboratory			
Application			
Field Work			
Study Hours Out of Class	7	2	14
Special Course Internship (Work Placement)			

Homework Assignments	8	2	16
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	2	2
Final (Examination Duration + Examination Prep. Duration)	1	2	2
		Total Workload	60
		Total Workload / 30(h)	2.00
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Turkish language 2	TDB1032	0	2	2	0	0

Prerequisite	None

Semester	Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	General Cultural Courses
Mode Of Delivery	Distance Learning

Owner Academic Unit	Department of Turkish Language
Course Coordinator	Yakup Çelik
Instructor(s)	Hilal TUFAN, Arzuhan KOCABAŞ, Zeliha ÇELEN BOZTİLKİ, Nurullah ARVAS
Asistant(s)	

Course Objectives	Correct use of Turkish, reading the professional and extraprofessional texts, successful oral and written expression.
Course Content	Reading sample literary and contemporary texts. Oral and written expression.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Know about the languages used in the world and the place of Turkish among world languages.
2	Can express themselves better by internalizing Turkish language and get recognition in the society.
3	Can understand and use their mother tongue in a better way.
4	Can use science and knowledge in a better way.
5	Can express themselves better in written and oral expression.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to written genres.	Textbook
2	Text analysis: Poem.	Contemporary Books
3	Text analysis: Story.	Contemporary Books
4	Text analysis: Fiction.	Contemporary Books
5	Problems with spelling of foreign origin words in Turkish accompanied by texts.	Contemporary Books
6	Genres dealing with the daily life (essay).	Textbook
7	Samples of written genres (Biography, autobiography).	Textbook
8	Midterm 1 / Practice or Review	
9	Oral expression.	Textbook
10	Correction of expression disorders, exercises.	Textbook

11	CV.	Textbook
12	Punctuation exercises.	Contemporary Books
13	The rules need to be considered in the prepared speech. Rules of pronunciation, style and the tone in public.	Book, Web
14	Presantation exercises.	Book, Web
15	Final	
16	Final exam.	Textbook

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury	1	30
Project		
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	2	28
Laboratory			
Application			
Field Work			
Study Hours Out of Class	7	2	14
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			
Presentations / Seminar	7	2	14
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	2	2

Final (Examination Duration + Examination Prep. Duration)	1	2	2
		Total Workload	60
		Total Workload / 30(h)	2.00
		ECTS Credit	2

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Expert Systems	BLM3760	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Fatih Amasyalı
Instructor(s)	M. Fatih Amasyalı
Asistant(s)	

Course Objectives	In this course, students will learn the concept of expert systems and, how to design an expert system.
Course Content	Basic concepts; Inference engine; Knowledge base; Knowledge elicitation; Representation and control of knowledge; Automated reasoning; Representing uncertainty; Practical problem solving; Development of the theory and practice of expert systems; Well known samples of expert systems; Software tools and architectures for building expert systems
Recommended Optional Program Components	None

Course Learning Outcomes

1	Implements an expert system.
2	Determines inference mechanism for a given problem
3	Determines knowledge representation method for a given problem
4	Knows the commonsense databases and their construction phases.
5	Knows the Prolog language at intermediate level.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction, History	
2	Basic concept: inference engine	
3	Knowledge Extraction, Semantic Web	
4	Commonsense Databases	
5	Prolog Language-1	
6	Prolog Language-2	
7	Prolog Language-3	
8	Midterm 1 / Practice or Review	

9	Prolog Language-4	
10	Knowledge Representation	
11	Case Base Reasoning	
12	Recomendation Systems	
13	Semantic Spaces-1	
14	Semantic Spaces-2	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments	2	25	50
Quizzes/Studio Critics			
Project	1	40	40
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	30	30
		Total Workload	238
		Total Workload / 30(h)	7.93
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Data Communication and Computer Networks	BLM3051	3	4	3	0	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Furkan Çakmak
Instructor(s)	Furkan Çakmak
Asistant(s)	

Course Objectives	To obtain a detailed view of the first four layers of OSI reference model, understand how LAN and WANs are operating and to have knowledge about data transmission technologies.
Course Content	OSI Reference Model; Signaling and Encoding Techniques; Serial and Parallel Transmission; Communication Media Specifications; Error Detection and Correction; Flow Control Techniques; Synchronous and Asynchronous Transmission; Connection Oriented and Connectionless Services and Their Specifications; Switching; Local Area Network Technologies; Wide Area Network Technologies; IP, TCP and UDP
Recommended Optional Program Components	None

Course Learning Outcomes

1	Knowledge on fundamentals of networks, network structures and network device's functions.
2	Learn signal encoding techniques and, knowledge on technical characteristics of various communication media.
3	Learn error detection techniques, flow control/management techniques and synchronous, asynchronous data link protocols.
4	Knowledge on local and wide area network technologies and ability to design computer networks.
5	Learn advanced knowledge of network layer features, routing, and network congestion.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Data Communication, Architectural Models	
2	OSI Referens Model, Layers and Their Functions	
3	Physical Layer and Signaling	
4	Parallel and Serial Transmission, Communication Media and Their Technical Specs., Multiplexing (TDM, FDM)	

5	Error Detection and Error Correction Techniques	
6	Data Link Control Techniques, Flow Control	
7	Synchronous and Asynchronous Data Link Protocols (BSC, HDLC)	
8	Midterm 1 / Practice or Review	Related reading form refence books
9	LAN Technologies, IEEE 802.3, 802.4, 802.5, 802.11	
10	Wide Area Network Technologies (X.25, ISDN, FR, ATM, xDSL)	
11	Connectionless and connection oriented services, Switching	
12	Static and Dynamic Routing, Network Layer Congestion, Causes and Solutions	
13	IP (Internetworking Protocol), ICMP, BOOTP, DHCP - Midterm 2	
14	UDP (User Datagram Protocol), TCP (Transmisson Control Protocol)	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury	1	10
Project	1	10
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28

Special Course Internship (Work Placement)			
Homework Assignments			0
Quizzes/Studio Critics			
Project	1	10	10
Presentations / Seminar	1	4	4
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	10	20
Final (Examination Duration + Examination Prep. Duration)	1	16	16
		Total Workload	120
		Total Workload / 30(h)	4.00
		ECTS Credit	4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Data Structures and Algorithms	BLM2512	4	6	3	0	2

Prerequisite	BLM1011 Introduction to Computer Sciences

Semester	Spring

Course Language	Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Elif Karslıgil
Instructor(s)	M. Elif Karslıgil, M. Amaç Güvensan, Göksel Biricik
Asistant(s)	

Course Objectives	The aim of this course is to teach the students to understand and implement fundemental data structures and algorithms and their effective use in a variety of applications.
Course Content	Fundamentals of Algorithmic Problem Solving, Fundamentals of the Analysis of Algorithm Efficiency, Lists and Linked Lists, Queues and Stacks, Trees, Graphs, Search Algorithms, Sorting Algorithms
Recommended Optional Program Components	None

Course Learning Outcomes

1	Student will understand how to design correct and efficient algorithms.
2	Student will learn major elementary data structures including stacks, queues, trees, graphs and should be able to use them appropriately to solve problems.
3	Student will learn the important algorithms and data structures in use on computers today
4	Student will learn the fundamendtal analysis and time complexity calculation methods of algorithms.
5	Student will able to apply prior knowledge of standard algorithms to solve new problems.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Fundamentals of Algorithmic Problem Solving	NA
2	Fundamentals of the Analysis of Algorithm Efficiency, Big-Oh Notation	NA
3	Lists and Linked Lists	NA
4	Queues and Stacks	NA
5	Trees, Binary Trees, Binary Search Trees	NA
6	Heap Trees	NA
7	Priority Queue Algorithms	NA

8	Midterm 1 / Practice or Review	NA
9	Presentation	NA
10	Minimum Spanning Trees	NA
11	String Search Algorithms	NA
12	Sorting Algorithms	NA
13	Presentation	NA
14	Shortest Path Algorithms	NA
15	Final	NA
16	Final Exam	NA

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		0
Laboratory	6	15
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	5
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	6	5	30
Application			
Field Work			
Study Hours Out of Class	15	4	60
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	15	15

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	10	20
Final (Examination Duration + Examination Prep. Duration)	1	10	10
		Total Workload	177
		Total Workload / 30(h)	5.90
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Database Management system Implementation	BLM3780	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Utku Kalay
Instructor(s)	M. Utku Kalay
Asistant(s)	

Course Objectives	Understanding DBMS implementation concepts with an educational DBMS prototype system.
Course Content	Buffer Management; Transaction Management; Concurrency Management; Recovery Management; Metadata Management; Query Execution; Query Planning; Cost-based Query Optimization
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student will have detailed information about the tasks and principles of data base system modules.
2	The student can understand implementation of a transactional database management system consisting of many modules such as buffer management and query execution and implementation.
3	Student can understand the main algorithms in the system implementation.
4	Students can propose new techniques in the area.
5	The student can make changes in some parts of the system, and analyze the effects of work.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Physical Storage Systems	Database Sys. Concepts. Chapter:12
2	Data Storage Structures	Database Sys. Concepts. Chapter:13
3	Indexing	Database Sys. Concepts. Chapter:14
4	Indexing	Database Sys. Concepts. Chapter:14
5	Query Processing	Database Sys. Concepts. Chapter:15

6	Query Processing	Database Sys. Concepts. Chapter:15
7	Query Optimization	Database Sys. Concepts. Chapter:16
8	Midterm 1 / Practice or Review	Database Systems and Implementation, Edward Sciori,,Ch:15,16
9	Query Optimization	Database Sys. Concepts. Chapter:16
10	Transactions	Database Sys. Concepts. Chapter:17
11	Concurrency Control	Database Sys. Concepts. Chapter:18
12	Concurrency Control	Database Sys. Concepts. Chapter:18
13	Recovery Systems	Database Sys. Concepts. Chapter:19
14	Recovery Systems	Database Sys. Concepts. Chapter:19
15	Final	Database Sys. Concepts. Chapter:25
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	1	10
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	12	6	72
Special Course Internship (Work Placement)			
Homework Assignments	1	20	20
Quizzes/Studio Critics			
Project	1	20	20
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	30	60
Final (Examination Duration + Examination Prep. Duration)	1	30	30
		Total Workload	244
		Total Workload / 30(h)	8.13
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Database Management	BLM3041	4	6	3	0	2

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Utku Kalay
Instructor(s)	M. Utku Kalay
Asistant(s)	Elçin GÜVEYİ

Course Objectives	Understanding database modeling, querying, and management techniques.
Course Content	Conceptual Design with ER/UML Modelling; Relational Model; Relational Algebra; SQL; DB Integrity Programming Techniques (Assertions, Triggers); DB-driven Programming Languages (Stored Procedures, Embedded SQL, JDBC); Normalization;Physical Design
Recommended Optional Program Components	None

Course Learning Outcomes

1	The student is able to design and model medium-scale databases.
2	The Student is able to criticize the database design issues and how to write basic SQL queries.
3	The student is able to use database management tools in the laboratory environment.
4	The student is able to design and program database-driven programs.
5	The student is able to apply new generation DB modelling/programming languages such as XML, XQuery and XPath

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to fundementals of database concepts and main architecture of db systems.	Ch 1
2	Introduction to the Relational Model	Ch 2
3	Intro to SQL	Ch 3
4	Intermediate SQL	Ch 4
5	Advanced SQL	Ch 5
6	Advanced SQL	Ch 5
7	Advanced SQL	Ch 5
8	Midterm 1 / Practice or Review	Ch 8,9

9	DB design- ER	Ch 6
10	DB design- ER	Ch 6
11	Normalization	Ch 7
12	Complex Data types	Ch 8
13	Application Development	Ch 9
14	Physical Design- B-tree- Hash	Ch 14
15	Final	Ch 14
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	12	20
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	10
Seminar/Workshop		
Mid-Terms	2	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	12	2	24
Application			
Field Work			
Study Hours Out of Class	13	6	78
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	14	14
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	2	5	10
Final (Examination Duration + Examination Prep. Duration)	1	15	15
		Total Workload	183
		Total Workload / 30(h)	6.10
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Neural Networks	BLM4520	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Sırma Yavuz
Instructor(s)	Sırma Yavuz
Asistant(s)	

Course Objectives	Learning basic problems and solution techniques in neural networks.
Course Content	Learning basic methods and applications used in neural networks. Determining whether a method fits into the problem given.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to recognize the basic problems in the aera.
2	Students will be able to construct the suitable models for the problems
3	Students will be able to determine the proper solution for the models
4	Students will understand the limitations of available tools
5	Students will know how to interpret results

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Why use neural networks ? Biological fondations of ANNs.	
2	Application areas, typical architectures, activation functions	
3	McCulloch-Pitts Neuron	
4	Simple Neural Networks for Pattern Classification	
5	Perceptron, Adaline, Delta Rule	
6	Multilayer Perceptrons	
7	Radial Based Networks	
8	Midterm 1 / Practice or Review	
9	Presentation	
10	Regression Analysis	
11	Learning Vector Quantization	

12	Pattern Association, learning algorithms, associative networks	
13	Pattern Association, learning algorithms, associative networks	
14	Hopfield Networks	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class			
Special Course Internship (Work Placement)			
Homework Assignments	2	35	70
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	40	80
Final (Examination Duration + Examination Prep. Duration)	1	40	40
		Total Workload	232

Total Workload / 30(h)	7.73
ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Artificial Intelligence	BLM3510	3	4	3	0	0

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	M. Fatih Amasyalı
Instructor(s)	M. Fatih Amasyalı
Asistant(s)	

Course Objectives	Gain the ability to problem solving with artificial intelligence algorithms.
Course Content	Learning artificial intelligence method, applications, and languages. Determining a problem is fit to AI methods or not.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Determines a problem is fit to AI methods or not.
2	Chooses an appropriate AI methods for a given problem.
3	Implements an AI methods for a given problem.
4	Knows the searching algorithms, their advantages and disadvantages.
5	Knows the knowledge representation methods, their advantages and disadvantages.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Course Introduction	
2	The history of AI	
3	Blind Search Algorithms	
4	Heuristic Search Algorithms	
5	Local Search Algorithms	
6	Genetic Algorithms	
7	Game Algorithms	
8	Midterm 1 / Practice or Review	
9	Knowledge Representation	
10	Machine Learning Algorithms-1	
11	Machine Learning Algorithms-2	

12	Reinforcement Learning	
13	Generative AI-1	
14	Generative AI-2	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	20
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	2	8	16
Quizzes/Studio Critics			
Project	1	12	12
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	128

Total Workload / 30(h)	4.27
ECTS Credit	4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Structured Programming	BLM1031	4	6	3	0	2

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Yunus Emre Selçuk
Instructor(s)	Yunus Emre Selçuk, H.İrem Türkmen
Asistant(s)	

Course Objectives	To teach advanced C programming
Course Content	Data Types ; Control Flows ; Loops ; Arrays and Pointers ; Multi-dimensional Arrays ; Pointer Arrays ; Strings ; Dynamic Memory Allocation ; Functions ; Function Pointers ; Recursive Functions ; Local and Global Variables ; Structures ; Bit-wise Operations ; File Operations ; C Preprocessing ; Macros ; Data Structures in C ; Linked Lists ; Static and Dynamic Libraries
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will have advanced C programming language skills.
2	Students will be able to design efficient algorithms.
3	Students will have modular programming skills.
4	Students will be able to use low level capabilities of C programming language.
5	Students will be able to write readable and reusable source codes.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Data Types / Control Flows / Loops	
2	Arrays / Pointers / Pointer Arithmetic	
3	Multi-dimensional Arrays / Pointer Arrays / Strings / Pointer to Pointers	
4	Dynamic Memory Allocation / Functions	
5	Function Pointers / Recursive Functions	
6	Local and Global Variables / Storage Classes	
7	Structs	
8	Midterm 1 / Practice or Review	

9	Applied Rehealsals	
10	Unions / Bit-wise Operations	
11	C Preprocessors and Macros	
12	Linked Lists	
13	I/O Operations	
14	Static and Dynamic Libraries	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	10	15
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	0	0
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	10	2	20
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	0	0	0
Quizzes/Studio Critics			
Project	1	32	32
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	40	40
		Total Workload	182
		Total Workload / 30(h)	6.07
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Software Quality and Testing	BLM4770	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oya Kalipsiz
Instructor(s)	Oya Kalipsiz
Asistant(s)	

Course Objectives	Teaching concepts related to software quality, quality assurance, software testing, software measurement and software verification & validation techniques
Course Content	Software Process and Product Ouality / Quality Assurance / Quality Engineering – Software Measurement and Metrics /Software Standards / Software Validation&Verification / Test Activities and Techniques / System Testing -Acceptence Test /Test case Design/Defect Prevention / Software Inspection /Software Reliability
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students know the quality management process and software quality standards.
2	Students know the distinctions between software verification and validation.
3	Students have the ability to run system and unit - component tests.
4	Students know how to prepare the software test suite.
5	Students have the ability to generate system test cases.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Software Quality Frameworks and– Quality Factors (ISO-9126 and- Mc Call,)	
2	Software Processing Models and Quality Approach	
3	Qualty Assurance Activities in Software Process	
4	Software Standards Software Process Improvement – Software Standards	
5	Quality Engineering and Software Metrics	
6	Software Verification & Validation	

7	Software Testing: Unit - Integrity - System Testing	
8	Midterm 1 / Practice or Review	
9	Software Testing: Test Automation	
10	Defect Preventetion, Software Inspection	
11	Interface and User Acceptence Testing	
12	Agile Software Development Models	
13	Agile Software Development Models and Test	
14	CMMI	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	2	30
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			

Project	2	47	94
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	18	18
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	230
		Total Workload / 30(h)	7.67
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Software Quality and Testing	BLM4770	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oya Kalipsiz
Instructor(s)	Oya Kalipsiz
Asistant(s)	

Course Objectives	Teaching concepts related to software quality, quality assurance, software testing, software measurement and software verification & validation techniques
Course Content	Software Process and Product Ouality / Quality Assurance / Quality Engineering – Software Measurement and Metrics /Software Standards / Software Validation&Verification / Test Activities and Techniques / System Testing -Acceptence Test /Test case Design/Defect Prevention / Software Inspection /Software Reliability
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students know the quality management process and software quality standards.
2	Students know the distinctions between software verification and validation.
3	Students have the ability to run system and unit - component tests.
4	Students know how to prepare the software test suite.
5	Students have the ability to generate system test cases.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Software Quality Frameworks and– Quality Factors (ISO-9126 and- Mc Call,)	
2	Software Processing Models and Quality Approach	
3	Qualty Assurance Activities in Software Process	
4	Software Standards Software Process Improvement – Software Standards	
5	Quality Engineering and Software Metrics	
6	Software Verification & Validation	

7	Software Testing: Unit - Integrity - System Testing	
8	Midterm 1 / Practice or Review	
9	Software Testing: Test Automation	
10	Defect Preventetion, Software Inspection	
11	Interface and User Acceptence Testing	
12	Agile Software Development Models	
13	Agile Software Development Models and Test	
14	CMMI	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	2	30
Seminar/Workshop		
Mid-Terms	1	30
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			

Project	2	47	94
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	18	18
Final (Examination Duration + Examination Prep. Duration)	1	20	20
		Total Workload	230
		Total Workload / 30(h)	7.67
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Software Engineering	BLM3722	3	4	3	0	0

Prerequisite	BLM2042 System Analysis and Design

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Yunus Emre Selçuk
Instructor(s)	Oya Kalipsiz, Yunus Emre Selçuk
Asistant(s)	

Course Objectives	Teaching the processes and methods of implementing high-quality and economic software
Course Content	Traditional Software Development Lifecycle Models; Agile Software Development Lifecycle Models; Requirements Engineering; Use-case Scenarios; UML Use-Case and Activity Diagrams; Software Testing; Software Quality; Software Maintenance; Software Re-use; Software Configuration Management; Software Project Management; Software Metrics and Measurement; Software Effort Estimation; Software Risk Management; Software Process Improvement (CMMI)
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will learn the classical and recent software development processes, including their advantages, disadvantages and appropriateness for different types of projects.
2	Students will learn the risks related with software development projects and gain the ability to carry out risk management tasks.
3	Students will gain the necessary prerequisites of working as a member or as the lead of a software development team.
4	Students will gain the ability to work in all phases of a software development project.
5	Students will gain the ability to create the technical documentation of a software development project.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to Software Engineering, Software Development Life Cycle Models (Traditional and Agile Models)	NA
2	Software Requirement Engineering. Use-case Scenarios. UML Use-Case diagrams.	NA
3	Introduction to Software Architecture	NA

4	Object Oriented Analysis and Design. UML Activity Diagrams.	NA
5	UML Modeling Tools	NA
6	Lab Activity: Analysis with UML modeling tools	NA
7	Software Testing.	NA
8	Midterm 1 / Practice or Review	NA
9	Software Configuration Management.	
10	Software Maintenance.	NA
11	Lab Activity: Design with UML modeling tools	NA
12	Software Project Management (Software Measurement and Effort Estimation Techniques, Agile Project Management)	NA
13	Software Process Improvement (CMMI) and Midterm Exam	
14	Presentation of the Students' Term Projects	NA
15	Final	
16	Final Exam	NA

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory	2	10
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	15
Seminar/Workshop		
Mid-Terms	2	35
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory	2	2	4
Application			
Field Work			

Study Hours Out of Class	15	2	30
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project	1	15	15
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	10	10
Final (Examination Duration + Examination Prep. Duration)	1	18	18
		Total Workload	119
		Total Workload / 30(h)	3.97
		ECTS Credit	4

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Management Information System	BLM4710	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oya Kalipsiz
Instructor(s)	Oya Kalipsiz
Asistant(s)	

Course Objectives	Study of the role of information systems in business enterprises and their application areas and ERP.
Course Content	Management information system (MIS) and other information systems / Management and Decision Processes / Information System Types Case Studies /ERP (Enterprise Resource Planning)/CRM, SCM and HR Systems /Upper Management Information Systems and Data Warehouses /Information Systems and Reorganisation in Business Enterprises / Total Quality Management and Information Systems
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students know how to plan for the use of information technologies.
2	Students develop their managerial skills in information systems.
3	Students describe information technology strategy and explain its alignment with organizational strategy.
4	Students describe ERP, SCM, CRM HR sistems concepts
5	Students know and describe how technology can provide competitive advantages to an organization.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Management and Decision Processes	
2	Information Systems and Reorganisation in Business Enterprises	
3	Decision Support Systems	
4	Enterprise Requirements Planning (ERP)	
5	CRM, SCM and HR Systems	
6	Executive Information Systems,	
7	Information System Types Case Studies	

8	Midterm 1 / Practice or Review	
9	Information Society Ethics	
10	Business Intelligence	
11	Total Quality Management and Information Systems	
12	Organizing Information Resources	
13	Impacts of Information Technology on Organizations	
14	Information Tech Project Management	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	10
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	14	4	56
Special Course Internship (Work Placement)			
Homework Assignments	2	18	36
Quizzes/Studio Critics			
Project	1	55	55

Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	20	20
Total Workload			229
Total Workload / 30(h)			7.63
ECTS Credit			8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Low Level Programming	BLM2021	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Furkan Çakmak
Instructor(s)	Furkan Çakmak, Erkan Uslu
Asistant(s)	

Course Objectives	Ability to use 80x86 assembly language as a low level programming language, to interact with Input-Output devices and interact with high level programming languages.
Course Content	Intel 8086 Processor Family Architecture; Registers and Their Functions; Flags;, 80x86 Assembly Mnemonics; Addressing Models; Pseudo Commands; EXE and COM Style Programs; Procedures and Procedure Calls; Macro; Segment Union; Parameter Passing Techniques; Interrupts; Interaction with High Level Programming Languages
Recommended Optional Program Components	None

Course Learning Outcomes

1	Defines registers of the 80x86 processor and their usage characteristics.
2	Knows the 80x86 processor's command set in real mode.
3	Describes the structural differences between EXE and COM type programs.
4	Being familiar with procedure calls, interrupt and macro handling as well as the methods of passing parameters.
5	Ability to write appropriate assembly programs, to debug errors and, if necessary, the use of these programs in conjunction with high-level languages.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	General specifications of low level programming language, Numbers, number systems and information coding systems. General introduction to 80x86 processor family, registers sets, flags and segmented organization	Text Book Chapter-1,2,3,4
2	Mnemonics (data transfer, arithmetic, branch)	Text Book Chapter-5
3	Mnemonics (loop, flags , logic , shift , rotate)	Text Book Chapter-5

4	Mnemonics (string operations, prefixes)	Text Book Chapter-5
5	Addressing modes, tools for low level programming, pseudo commands	Text Book Chapter-6, 7, 8
6	Setting up the programming environment, debug tools	Text Book Chapter-16
7	EXE style programming	Text Book Chapter-9
8	Midterm 1 / Practice or Review	
9	COM style programming	Text Book Chapter-9
10	Procedures and Macros	Text Book Chapter-10
11	Sub-procedures and parameter passing methods	Text Book Chapter-11, 12
12	Common segment usage EXTRN and PUBLIC definitions	Text Book Chapter-11, 12
13	Exercises	
14	Interrupts, vector table and integration with high level programming languages	Text Book Chapter-13, 14, 15
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			

Study Hours Out of Class	14	3	42
Special Course Internship (Work Placement)			
Homework Assignments	2	24	48
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	24	48
Final (Examination Duration + Examination Prep. Duration)	1	60	60
		Total Workload	240
		Total Workload / 30(h)	8.00
		ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Object Oriented Programming	BLM2012	4	6	3	0	2

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Yunus Emre Selçuk
Instructor(s)	Yunus Emre Selçuk, Mehmet Sıddık Aktaş, Furkan Çakmak
Asistant(s)	

Course Objectives	Become able to design and implement object-oriented code by using Java and UML
Course Content	Objects, Classes and Members; Final and Static Members; Constructors and Finalizers; UML Class Diagrams; Command-line I/O; Control Flow; Relationships Between Classes and Objects (Association, Dependency, Aggregation, Composition, Inheritance); Overriding and Overloading; Primitives and Wrappers; Enum; Exception Handling; File Operations (Serialization and Deserialization using Streams); Generics; List and Map Data Structures; Introduction to Multithreading;
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to do object oriented modeling for the business logic layer of an information system.
2	Students will be able to document their designs using UML Class and Sequence diagrams
3	Students will be able to make two-way transformation between Java code and learned UML diagrams
4	Students will be able to write Java code that works from the command prompt
5	Students will be able to carry out fundamental tasks with modern IDE's

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction to the course and the Java language	
2	Classes, objects, members. Special cases: final, static. UML Class diagrams	
3	Constructors and finalizers. Control flow. Creating objects.	
4	UML Sequence diagrams. Constructor and method overloading. Primitives. String and Math classes. Command line I/O.	
5	Association and dependency. One-way and two-way association.	

6	Aggregation and Composition.	
7	Inheritance. Overriding and overloading.	
8	Midterm 1 / Practice or Review	
9	Abstract Classes and Interfaces	
10	Primitives, wrappers, parameters.	
11	Exception handling.	
12	Working with Files and Streams (Serialization and deserialization).	
13	Introduction to generic classes using basic data structures (Lists and Maps).	
14	Typecasting, Enum classes, Inner classes.	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	15	
Laboratory	4	10
Application	4	
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project	1	25
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	15	3	45
Laboratory	4	2	8
Application	4	2	8
Field Work			
Study Hours Out of Class	15	2	30
Special Course Internship (Work Placement)			

Homework Assignments			
Quizzes/Studio Critics			
Project	1	30	30
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	30	30
		Total Workload	171
		Total Workload / 30(h)	5.70
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Multidisciplinary Design Project	KOM4991	1	3	0	2	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Control and Automation Engineering
Course Coordinator	Şeref Naci Engin
Instructor(s)	Şeref Naci Engin, Veysel Gazi, İbrahim Alışkan, Levent Ucun, Yavuz Eren, Muharrem Mercimek, Claudia Fernanda Yaşar, Uğur Yıldıran, Fatih Adıgüzel, Bahadır Çatalbaş
Asistant(s)	

Course Objectives	Effective Teamwork and Communication, In Conceptual Project Management and Engineering Design, Understanding and Identification of a given engineering problem, Continuous Learning Requirement and develop the skills to absorb the Rules of Ethics
Course Content	Conceptual Preparation Phase: Design and Project Concept / Team Work and Project Management / Design Tools and Methods / Statistics and Modelling Methods /Environmental and Social Interaction / Flexibility and Reliability / Quality and Standard Concepts / Price and Engineering Economics / Ethics and Concepts / Reporting and Presentation Techniques / Reporting and Presentation Örneklerırımın Tools and Methods. Team Project Work 1: Determination of Issues Project / Project Teams Building / Planning and the Division of Labour / Resource Research and Literature / Resources Classification and Analysis / Determination of tools and methods to be used / Analysis and Modeling / Design and Verification / Procurement and Implementation / Testing and Testing / Analysis and Interpretation of Results / The Problems and Solutions / Quality and Value Assessment / Environmental and Social Assessment / Reporting and Presentation
Recommended Optional Program Components	None

Course Learning Outcomes

1	Ability to Work Between Disciplines
2	Ability to Transfer Project Outcomes in Oral and Verbal
3	The Need for Continuous Learning and Assimilation of Ethical Rules
4	Project Management and Engineering Design
5	Understanding and Defining a Given Engineering Problem and To Define and Utilize the Techniques, Tools and IT Requirements

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Determination of Project Issues	
2	Creating Project Teams	
3	Planning and Division of Labour	
4	Resource Research and Literature / Resources Classification and Analysis	
5	Determination of tools and methods to be used	
6	Analysis and Modeling	
7	Design and Verification	
8	Midterm 1 / Practice or Review	
9	Procurement and Implementation	
10	Tests and Experiments	
11	Tests and Experiments	
12	Analysis and Interpretation of Results	
13	Identification and solution of problems	
14	Reporting and Presentation	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation	13	50
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	10
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload

Course Hours			0
Laboratory			
Application	13	2	26
Field Work			
Study Hours Out of Class	13	3	39
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			0
Presentations / Seminar			0
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	8	8
Final (Examination Duration + Examination Prep. Duration)	1	8	8
		Total Workload	81
		Total Workload / 30(h)	2.70
		ECTS Credit	3

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Statistics and Probability	BLM2011	3	6	3	0	0

Prerequisite	None

Semester	Fall

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Oğuz Altun
Instructor(s)	Oğuz Altun, Ayşe Öcal
Asistant(s)	

Course Objectives	Learning how to model and solve problems involving uncertainty using probabilistic and statistical models and methods.
Course Content	Elementary probability rules, discrete and continuous probabilistic models, descriptive statistics, elementary statistical inference
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students calculate the statistics of numerical data.
2	Students use concepts of probability, and random variables to represent uncertainty within phenomena.
3	Students can solve probability computation problems involving probability density and cumulative distribution functions.
4	Students can recognize several common probability distributions and solve distribution problems.
5	Students can construct confidence intervals and conduct hypothesis test.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Introduction	Baron Ch. 1
2	Events and probabilities, rules of probability	Baron Ch. 2
3	Discrete random variables	Baron Ch. 3
4	Discrete random variables	Baron Ch. 3
5	Continuous random variables	Baron Ch. 4
6	Continuous random variables	Baron Ch. 4
7	Summary statistics	Baron Ch. 8
8	Midterm 1 / Practice or Review	Baron Chapter 8
9	Graphical statistics	Baron Ch. 8

10	Parameter estimation	Baron Ch. 9.1
11	Confidence Intervals	Baron Ch. 9.2
12	Unkown Standard Deviation	Baron Ch. 9.3
13	Hypothesis Testing	Baron Ch. 9.4
14	Inference About Variances	Baron Ch. 9.5
15	Final	Baron Ch. 9
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	60
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	25	50

Final (Examination Duration + Examination Prep. Duration)	1	39	39
		Total Workload	183
		Total Workload / 30(h)	6.10
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction to Neural Networks	BLM4520	3	8	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Sırma Yavuz
Instructor(s)	Sırma Yavuz
Asistant(s)	

Course Objectives	Learning basic problems and solution techniques in neural networks.
Course Content	Learning basic methods and applications used in neural networks. Determining whether a method fits into the problem given.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will be able to recognize the basic problems in the aera.
2	Students will be able to construct the suitable models for the problems
3	Students will be able to determine the proper solution for the models
4	Students will understand the limitations of available tools
5	Students will know how to interpret results

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Why use neural networks ? Biological fondations of ANNs.	
2	Application areas, typical architectures, activation functions	
3	McCulloch-Pitts Neuron	
4	Simple Neural Networks for Pattern Classification	
5	Perceptron, Adaline, Delta Rule	
6	Multilayer Perceptrons	
7	Radial Based Networks	
8	Midterm 1 / Practice or Review	
9	Presentation	
10	Regression Analysis	
11	Learning Vector Quantization	

12	Pattern Association, learning algorithms, associative networks	
13	Pattern Association, learning algorithms, associative networks	
14	Hopfield Networks	
15	Final	
16		

Evaluation System

Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	2	20
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	2	40
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours	14	3	42
Laboratory			
Application			
Field Work			
Study Hours Out of Class			
Special Course Internship (Work Placement)			
Homework Assignments	2	35	70
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	2	40	80
Final (Examination Duration + Examination Prep. Duration)	1	40	40
		Total Workload	232

Total Workload / 30(h)	7.73
ECTS Credit	8

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Structured Programming	BLM1031	4	6	3	0	2

Prerequisite	None

Semester	Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Yunus Emre Selçuk
Instructor(s)	Yunus Emre Selçuk, H.İrem Türkmen
Asistant(s)	

Course Objectives	To teach advanced C programming
Course Content	Data Types ; Control Flows ; Loops ; Arrays and Pointers ; Multi-dimensional Arrays ; Pointer Arrays ; Strings ; Dynamic Memory Allocation ; Functions ; Function Pointers ; Recursive Functions ; Local and Global Variables ; Structures ; Bit-wise Operations ; File Operations ; C Preprocessing ; Macros ; Data Structures in C ; Linked Lists ; Static and Dynamic Libraries
Recommended Optional Program Components	None

Course Learning Outcomes

1	Students will have advanced C programming language skills.
2	Students will be able to design efficient algorithms.
3	Students will have modular programming skills.
4	Students will be able to use low level capabilities of C programming language.
5	Students will be able to write readable and reusable source codes.

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Data Types / Control Flows / Loops	
2	Arrays / Pointers / Pointer Arithmetic	
3	Multi-dimensional Arrays / Pointer Arrays / Strings / Pointer to Pointers	
4	Dynamic Memory Allocation / Functions	
5	Function Pointers / Recursive Functions	
6	Local and Global Variables / Storage Classes	
7	Structs	
8	Midterm 1 / Practice or Review	

9	Applied Rehealsals	
10	Unions / Bit-wise Operations	
11	C Preprocessors and Macros	
12	Linked Lists	
13	I/O Operations	
14	Static and Dynamic Libraries	
15	Final	
16		

Evaluation System

Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory	10	15
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments	0	0
Presentations/Jury		
Project	1	20
Seminar/Workshop		
Mid-Terms	1	25
Final	1	40
	Percentage of In-Term Studies	60
	Percentage of Final Examination	40
	TOTAL	100

ECTS Workload Table

Activities	Number	Duration(Hour)	Total Workload
Course Hours	14	3	42
Laboratory	10	2	20
Application			
Field Work			
Study Hours Out of Class	14	2	28
Special Course Internship (Work Placement)			
Homework Assignments	0	0	0
Quizzes/Studio Critics			
Project	1	32	32
Presentations / Seminar			

Mid-Terms (Examination Duration + Examination Prep. Duration)	1	20	20
Final (Examination Duration + Examination Prep. Duration)	1	40	40
		Total Workload	182
		Total Workload / 30(h)	6.07
		ECTS Credit	6

Extra Notes	None

Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Introduction To Procedural Programming	BLM1012	3	5	3	0	0

Prerequisite	None

Semester	Fall, Spring

Course Language	English, Turkish
Level Of Course	First Cycle
Course Category	Major Area Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Computer Engineering
Course Coordinator	Not Assigned
Instructor(s)	
Asistant(s)	

Course Objectives	
Course Content	
Recommended Optional Program Components	None

Course Learning Outcomes

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1		
2		
3		
4		
5		
6		
7		
8	Midterm 1 / Practice or Review	
9		
10		
11		
12		
13		
14		
15	Final	
16		

Evaluation System		
Activities	**Number**	**Percentage of Grade**
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics		
Homework Assignments		
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms		
Final		
	Percentage of In-Term Studies	0
	Percentage of Final Examination	
	TOTAL	0

ECTS Workload Table			
Activities	**Number**	**Duration(Hour)**	**Total Workload**
Course Hours			
Laboratory			
Application			
Field Work			
Study Hours Out of Class			
Special Course Internship (Work Placement)			
Homework Assignments			
Quizzes/Studio Critics			
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)			
Final (Examination Duration + Examination Prep. Duration)			
		Total Workload	0
		Total Workload / 30(h)	0.00
		ECTS Credit	0

Extra Notes	None